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Network Logs Monitoring & Analysis – Windows Event Logs, IIS, Splunk SIEM & New Relic

Executive Summary

Logs never lie — if you know how to listen.

Module 15 transforms raw system noise into actionable security intelligence. In a clean VMware lab (VMnet3, 192.168.100.0/24), four machines work together: Windows Server 2019 as the log-rich target, Windows 10 as the analyst station, and Linux attackers generating real activity.

Students first unlock Windows Event Logs — enabling full Success/Failure auditing, creating Event IDs 4624 (successful logon) and 4625 (failed attempts), then dissecting them in Event Viewer and exporting forensic evidence. Next comes IIS web-server logging: precise field configuration, live HTTP traffic generation, and deep analysis of the resulting access logs.

The game changes with centralized power. Splunk Enterprise is installed on the Admin machine and fed by a Universal Forwarder on the Web Server, pulling both Windows Security events and IIS logs into a single searchable SIEM. Finally, New Relic brings modern observability — real-time host metrics, system logs, and instant detection of anomalies such as a sudden traffic spike from a ping scan.

Result: complete mastery of local log configuration, forensic analysis, centralized SIEM collection, and cloud-grade observability — the exact skills every defender needs to turn logs into early warnings.

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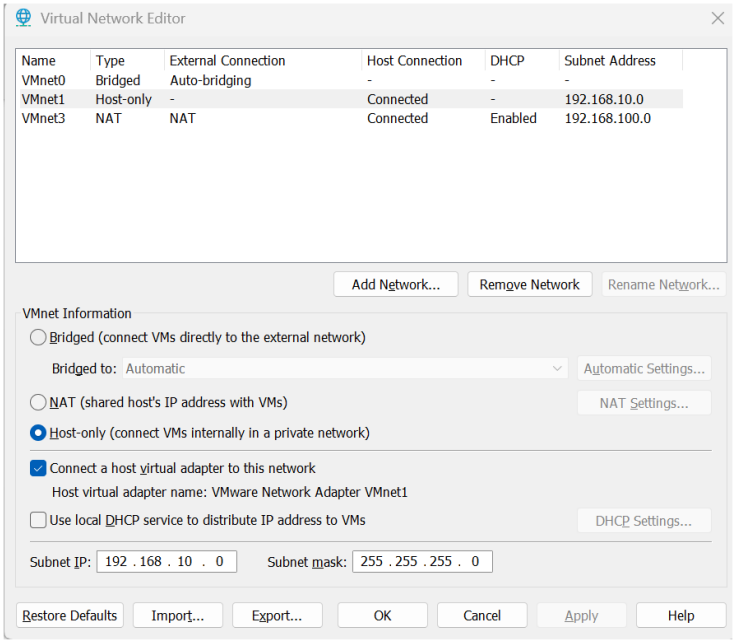
Generating Brute-Force Activity 55

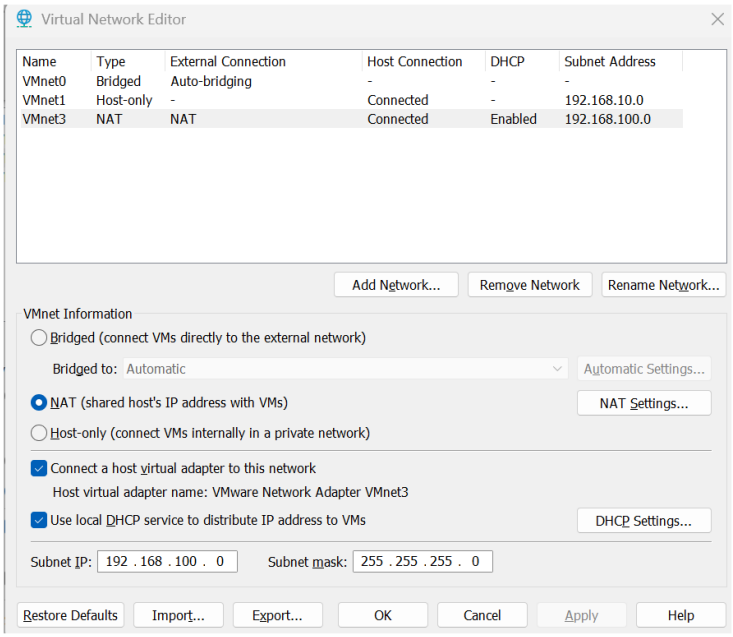
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Virtual Network Editor Setting

The VMware Virtual Network Editor is used to configure the virtual networks (VMnets) used by the lab VMs.

- **VMnet1** — Host-only network, Subnet: 192.168.10.0, connected to host, no DHCP.
- **VMnet3** — NAT network, Subnet: 192.168.100.0, DHCP enabled, connected to host.

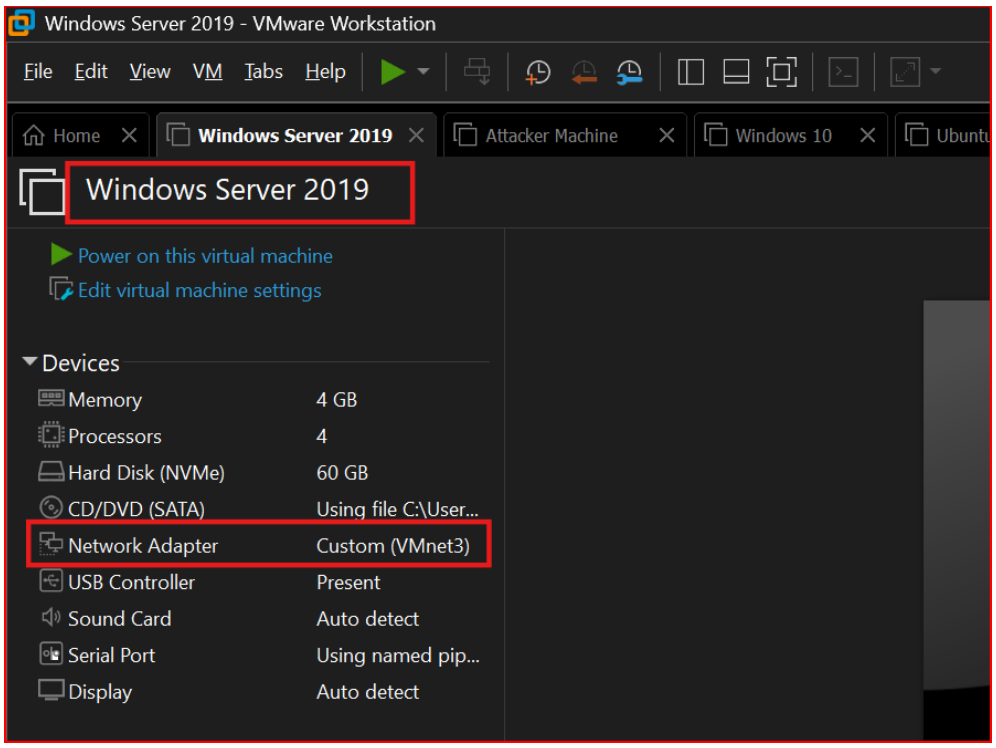




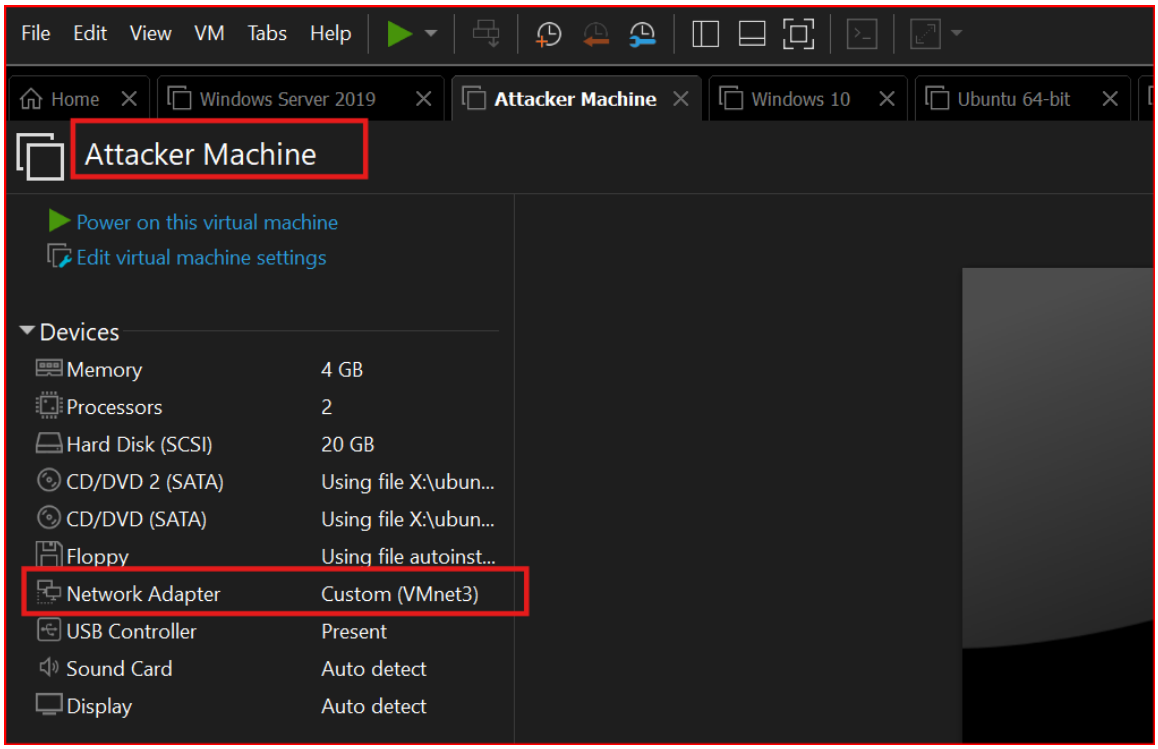
VM IP Address Table

VM Name	OS	Role in Module 15	Adapter	IP Address
Windows 10	Windows 10	Admin Machine-1 / Monitoring Server	VMnet3	**192.168.100.138**
Windows Server 2019	Windows Server 2019	Web Server + IIS Logs + Windows Event Logs	VMnet3	**192.168.100.130**
Attacker Machine	Linux Security VM	Attack simulation / suspicious activity generation	VMnet3	**192.168.100.143**
Ubuntu 64-bit	Ubuntu	Network traffic generation / monitoring test	VMnet3	**192.168.100.40**

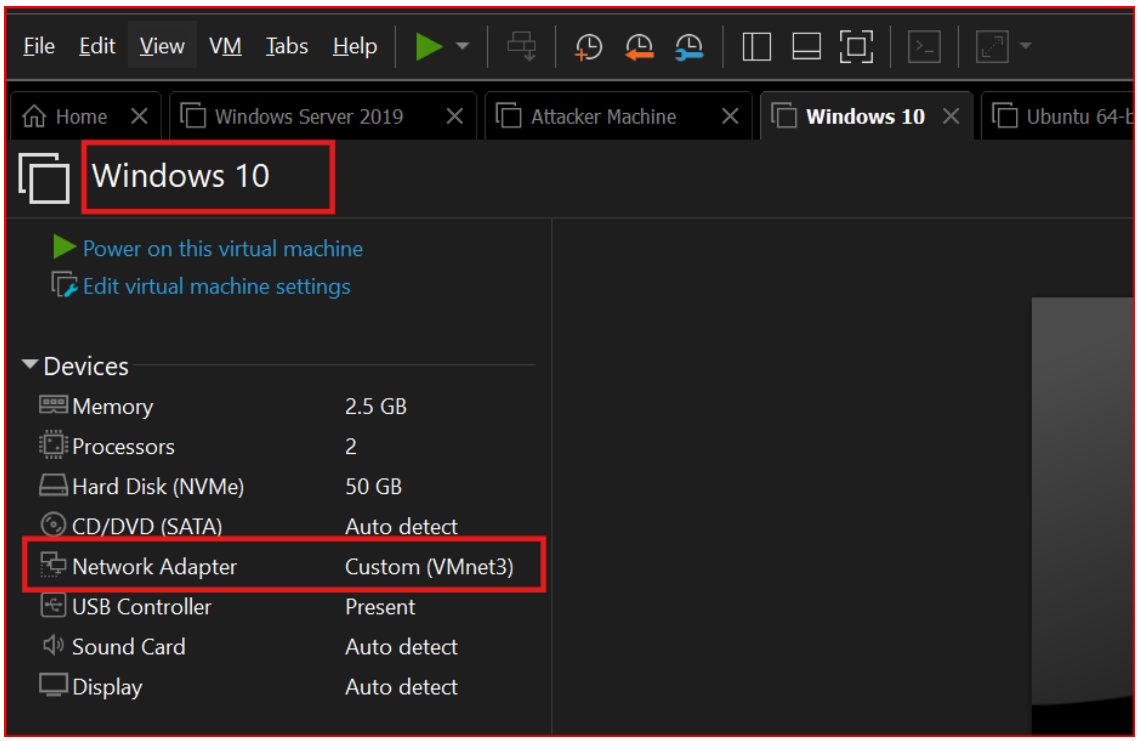
Web Server



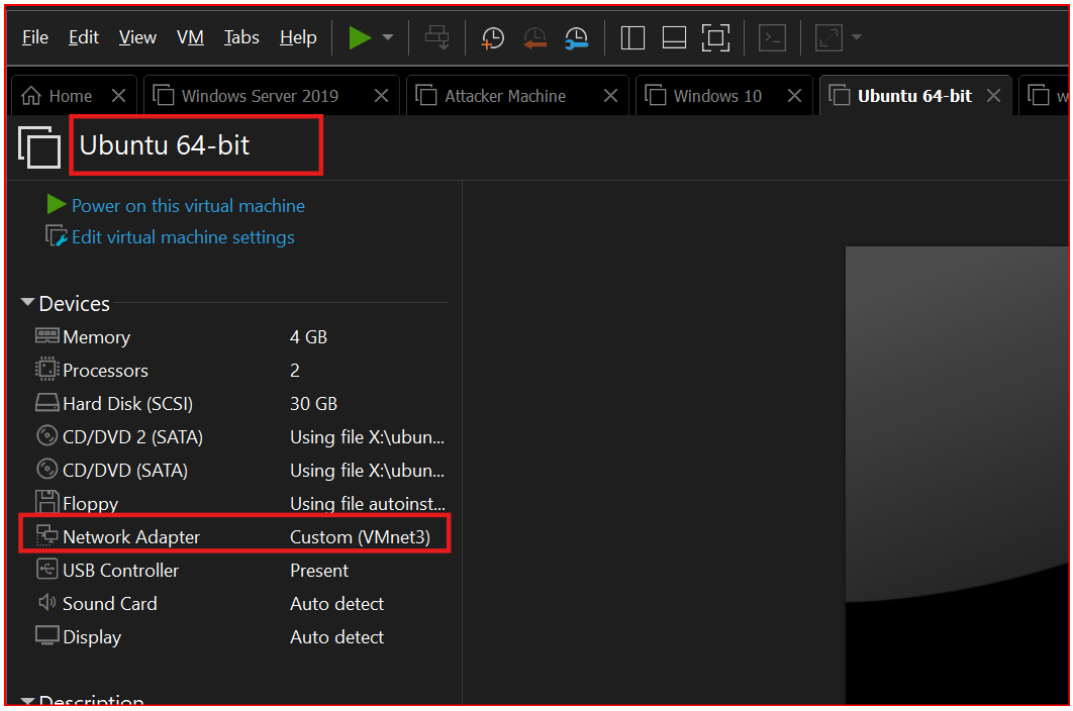
Attacker Machine (Ubuntu)



Windows Admin Machine



2nd Attacker Machine (Ubuntu)



Connectivity test

```
C:\Users\cyber>ping 192.168.100.130

Pinging 192.168.100.130 with 32 bytes of data:
Reply from 192.168.100.130: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.100.130:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
Control-C
^C
C:\Users\cyber>
C:\Users\cyber>ping 192.168.100.143

Pinging 192.168.100.143 with 32 bytes of data:
Reply from 192.168.100.143: bytes=32 time=4ms TTL=64
Reply from 192.168.100.143: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.100.143:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms
Control-C
```

Configuring, Viewing, and Analyzing Windows Event Logs

Configure Windows auditing, generate security events, and analyze Windows Event Logs to understand system activity and authentication events.

Objectives

Enable Windows auditing
Generate security events
Capture login-related events
Analyze Event Viewer logs
Identify important security event information
Understand how defenders investigate Windows activity

Machines Required

Machine	Purpose
Windows Server 2019	Main log generation machine
Windows 10	Admin/analysis machine
Attacker Machine	Generate suspicious activity

Ubuntu is **not required for Lab 1**.

Tools Required

Tool	Status
Windows Event Viewer	Built-in
Local Security Policy	Built-in
Command Prompt	Built-in

No external installation required.

Machine: Windows Server 2019 (Web Server)

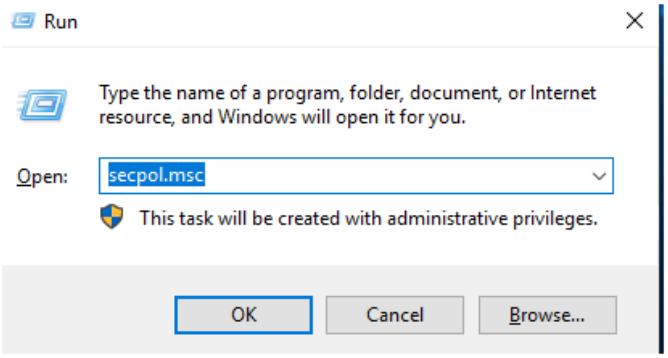
Configure Windows Audit Policy

Enable Windows Security Auditing on **Windows Server 2019** so the operating system records security activities.

Open Local Security Policy

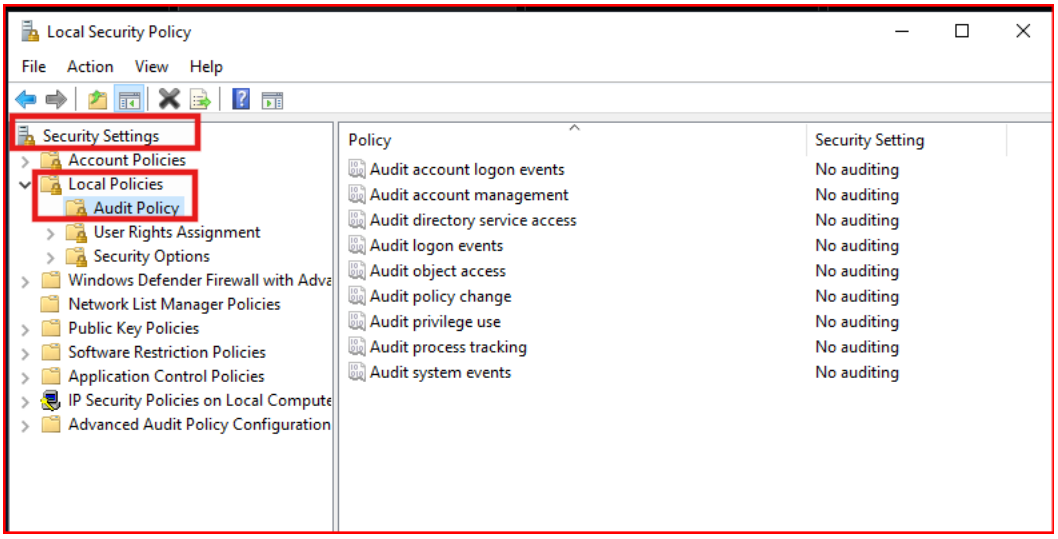
1. Press: Windows Key + R

Type: `secpol.msc`



Navigate to Audit Policy

Security Setting -> Local Policies ->Audit Policy

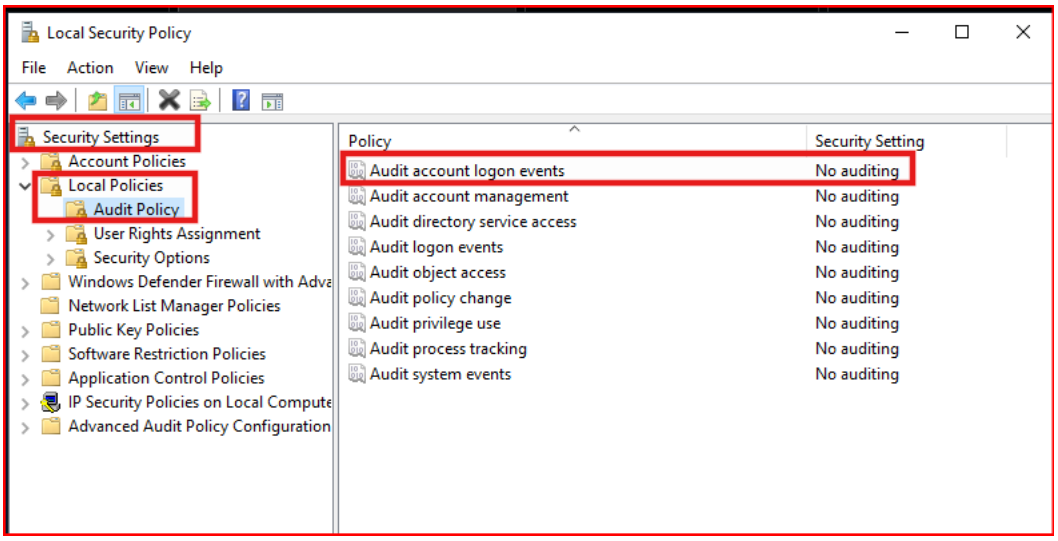


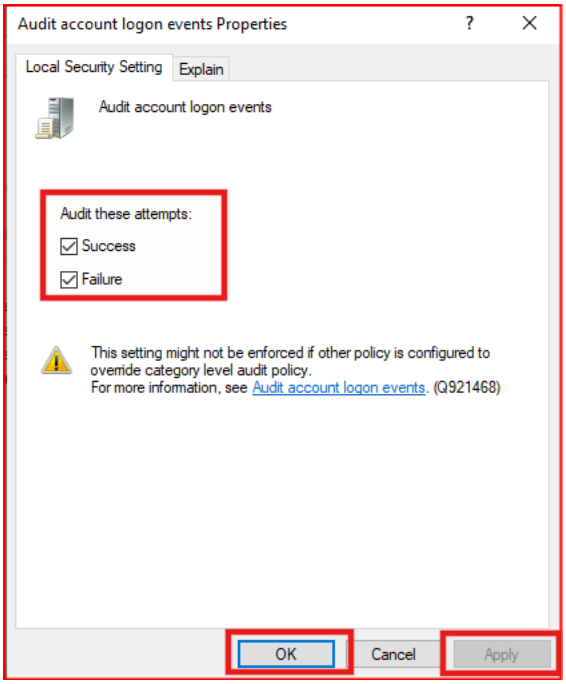
Enable Audit Logon Events

Audit logon events

Select:

- Success
- Failure



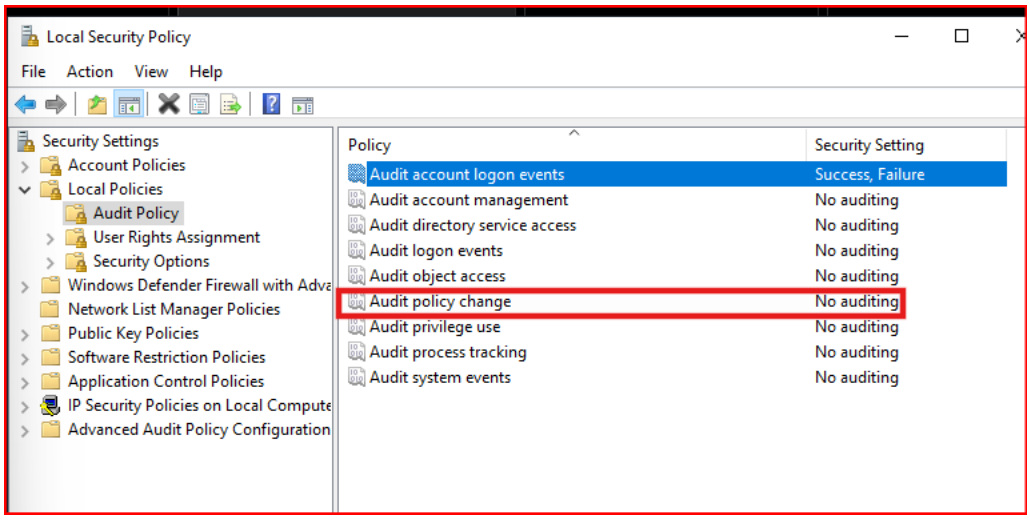


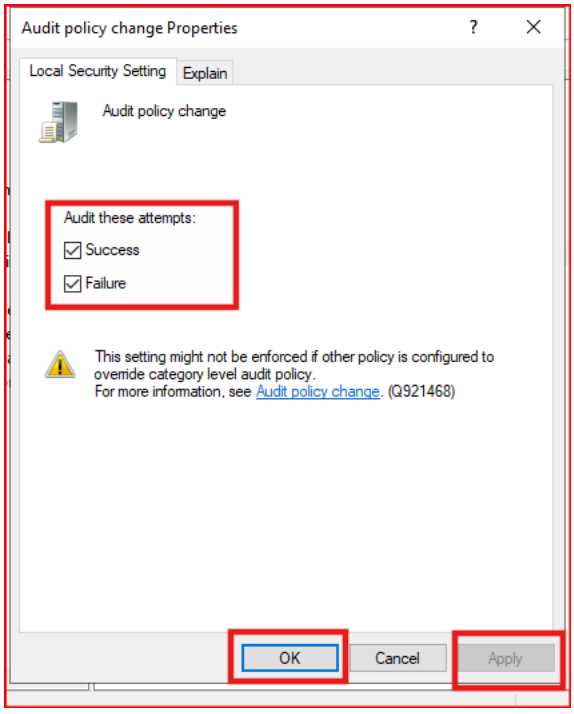
Enable Audit Policy Change

Audit policy change

Select:

- Success
- Failure





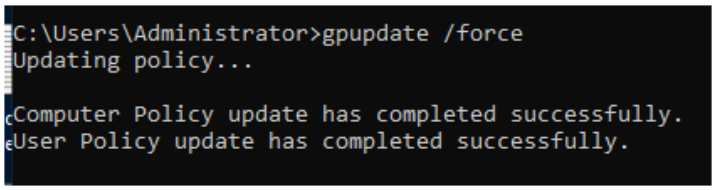
Force Policy Update

Open Command Prompt as Administrator.

Run: `gpupdate /force`

Purpose:

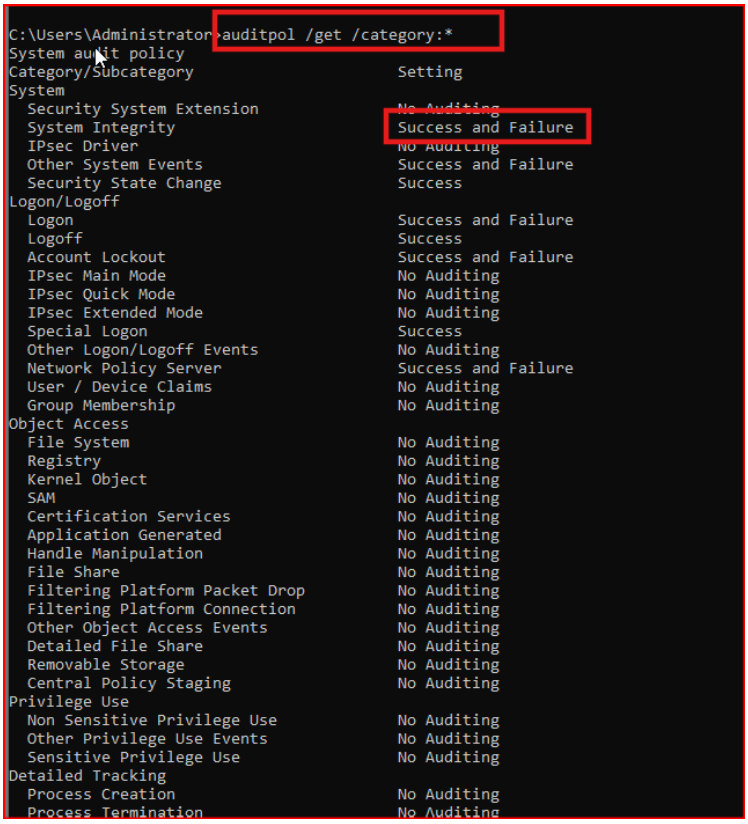
- Applies the new auditing configuration immediately
- Refreshes Windows Group Policy settings

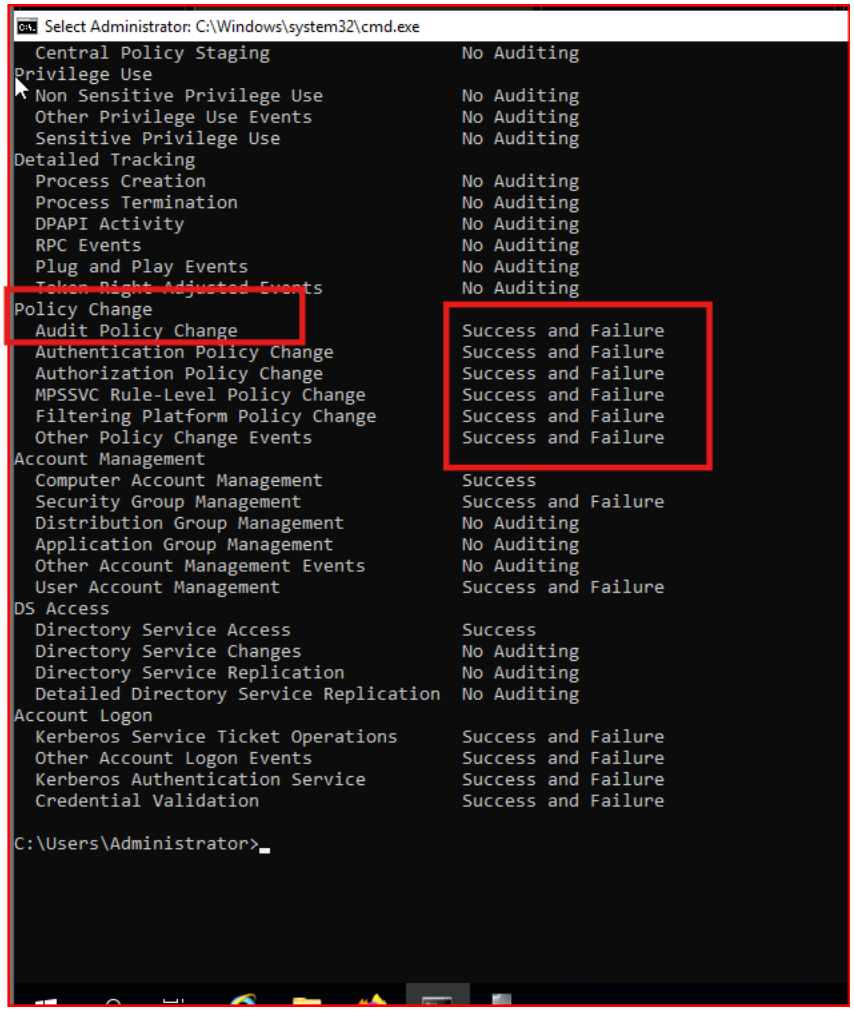


Verify Audit Configuration

Run: `auditpol /get /category:*`

Expected output should show enabled auditing like:




Select Administrator: C:\Windows\system32\cmd.exe
Central Policy Staging No Auditing
Privilege Use
Non Sensitive Privilege Use No Auditing
Other Privilege Use Events No Auditing
Sensitive Privilege Use No Auditing
Detailed Tracking
Process Creation No Auditing
Process Termination No Auditing
DPAPI Activity No Auditing
RPC Events No Auditing
Plug and Play Events No Auditing
Token Right Adjusted Events No Auditing
Policy Change
Audit Policy Change Success and Failure
Authentication Policy Change Success and Failure
Authorization Policy Change Success and Failure
MPSSVC Rule-Level Policy Change Success and Failure
Filtering Platform Policy Change Success and Failure
Other Policy Change Events Success and Failure
Account Management
Computer Account Management Success
Security Group Management Success and Failure
Distribution Group Management No Auditing
Application Group Management No Auditing
Other Account Management Events No Auditing
User Account Management Success and Failure
DS Access
Directory Service Access Success
Directory Service Changes No Auditing
Directory Service Replication No Auditing
Detailed Directory Service Replication No Auditing
Account Logon
Kerberos Service Ticket Operations Success and Failure
Other Account Logon Events Success and Failure
Kerberos Authentication Service Success and Failure
Credential Validation Success and Failure
C:\Users\Administrator>

Generate Security Events

Create security events manually so Windows Event Viewer has records to analyze.

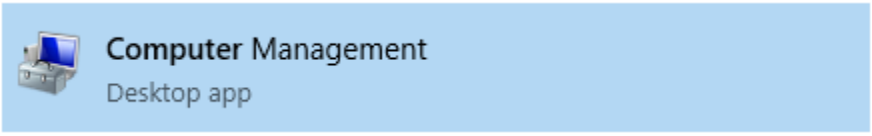
We will generate:

1. **Failed login attempts**
    - Creates Event ID **4625**
    - Shows unsuccessful authentication
2. **Successful login activity**
    - Creates Event ID **4624**
    - Shows successful authentication

These events will later be analyzed in the coming phase…

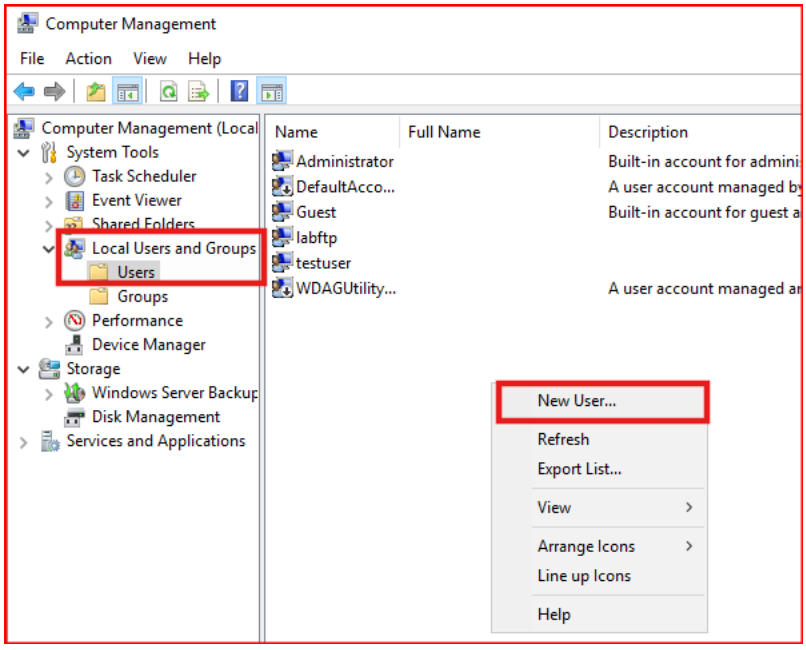
Create Test User on Windows Server

Computer Management



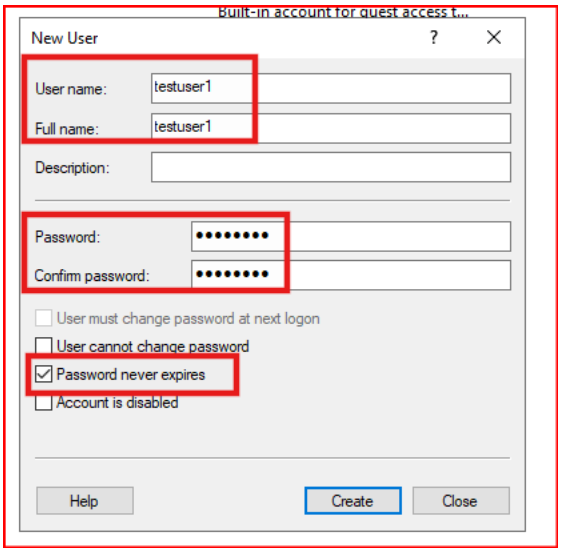
Navigate:

Local Users and Groups -> Users



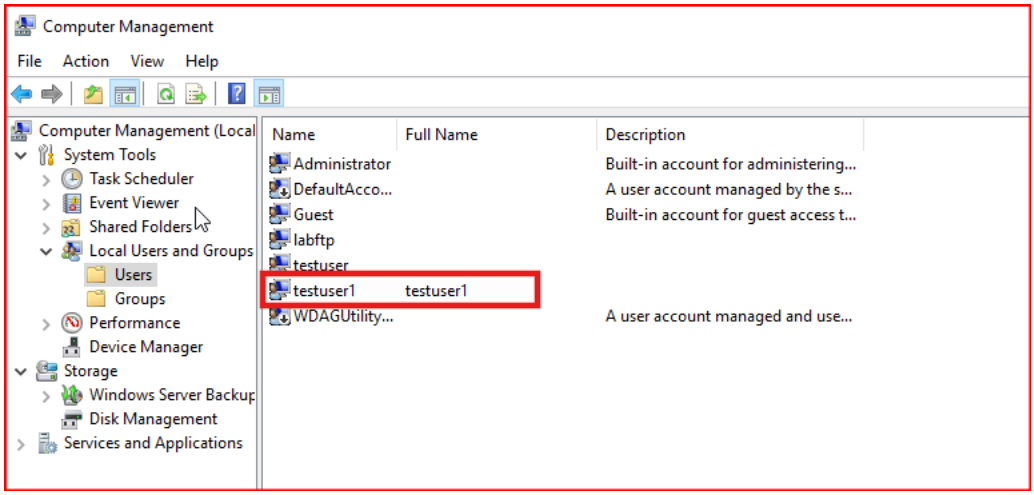
Create a new user:

- Username: testuser1
- Password: Admin123
- Enable: Password never expires



Purpose:

- Creates a controlled account for testing
- Avoids affecting administrator accounts



Machine: Attacker Machine

Generate Failed Login Events

Run:

- ssh testuser1@192.168.100.130
- Enter wrong password

Repeat this:

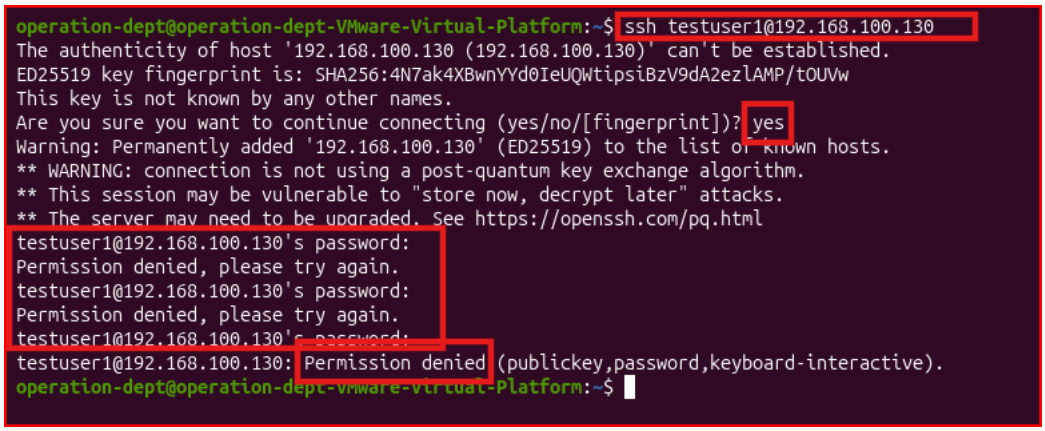
3-5 times

Expected:

Permission denied

Purpose:

- Creates failed authentication events
- Windows Server records Event ID 4625



Generate Successful Login Event

ssh testuser1@192.168.100.130

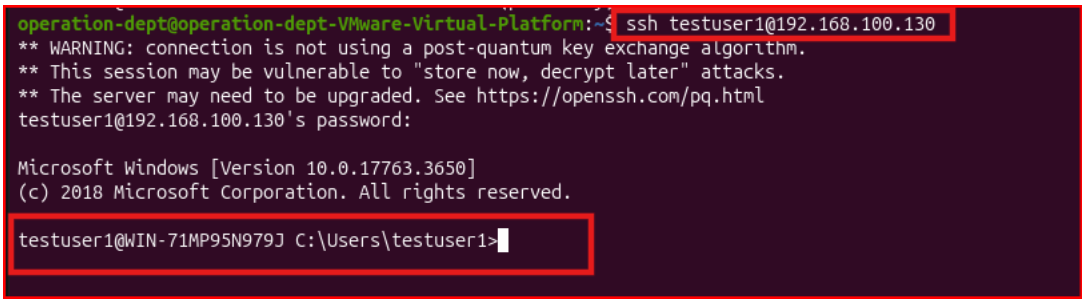
Enter correct password: Admin123

Expected:

Successful login

Purpose:

- Creates Event ID 4624
- Shows successful authentication



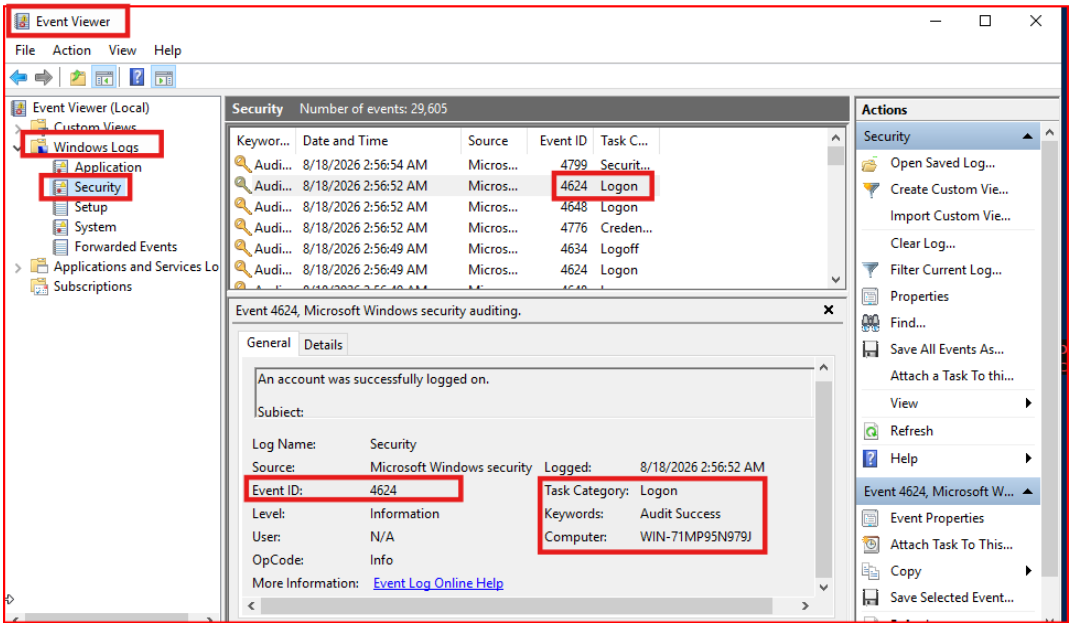
Analyze Windows Security Events

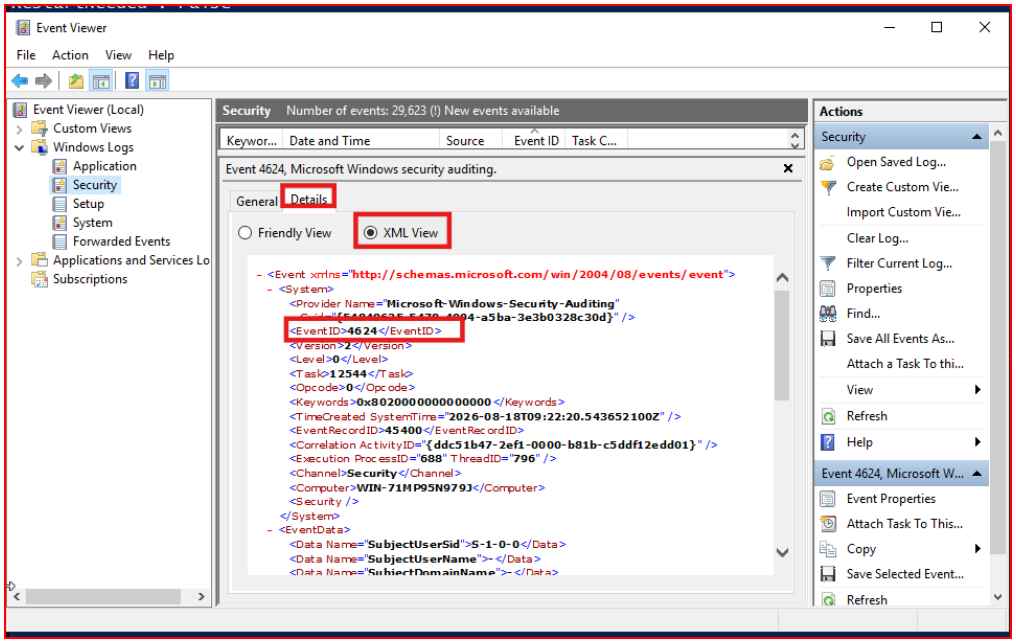
Open: Event Viewer

Navigate:Windows Logs ->Security

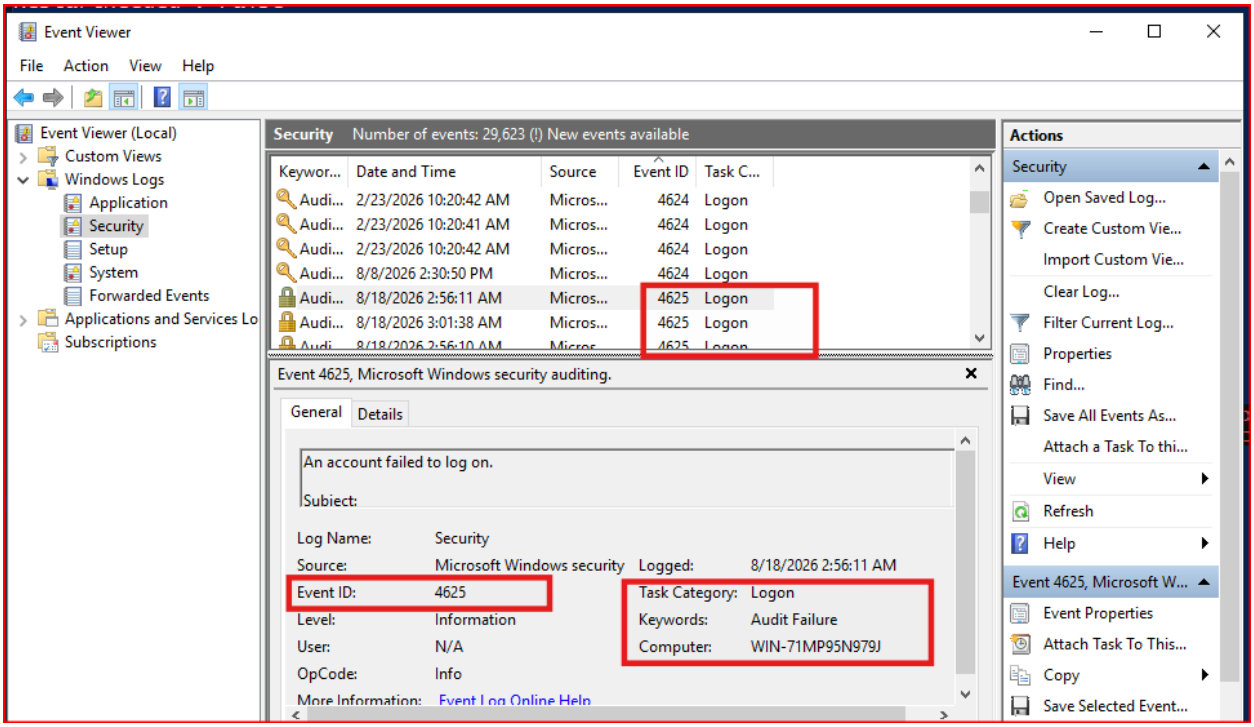
Click: Filter Current Log

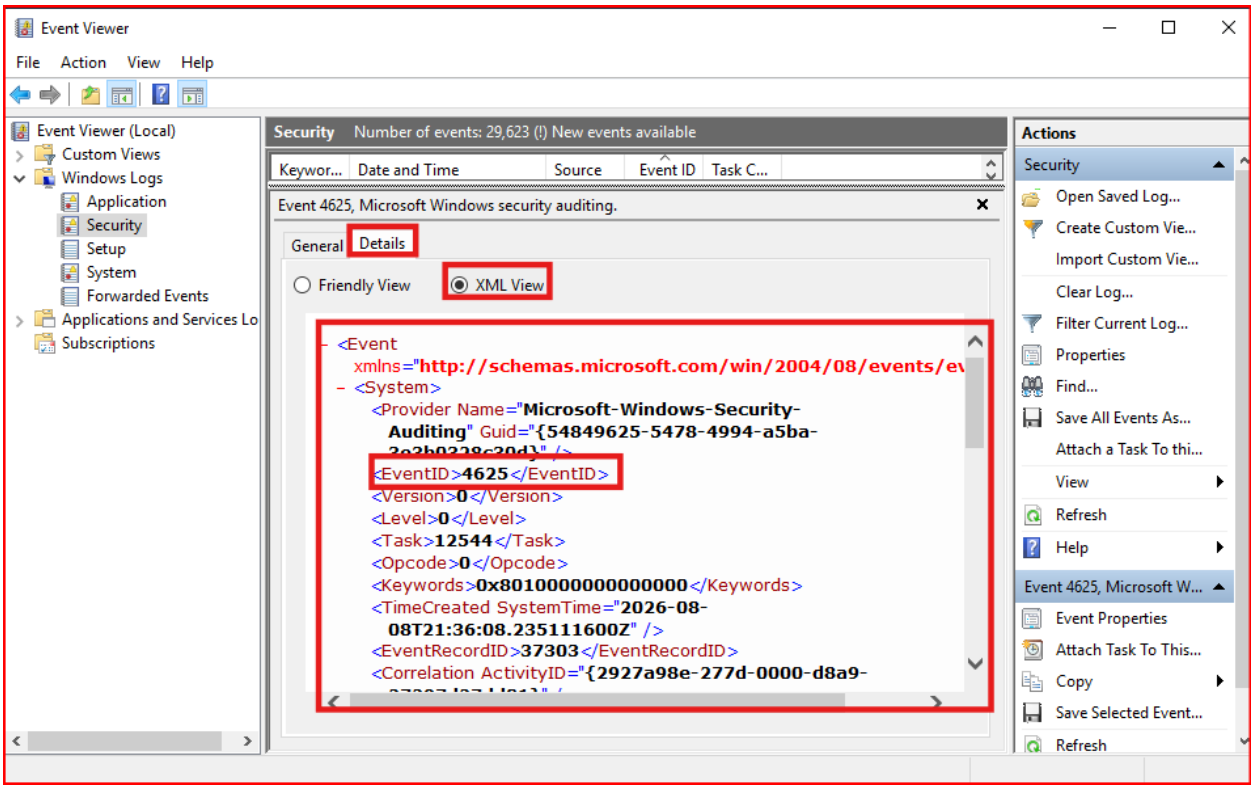
Analyze Successful Login Event (4624)





Analyze Failed Login Event (4625)





Export Security Log Evidence

In Event Viewer:

Right-click:

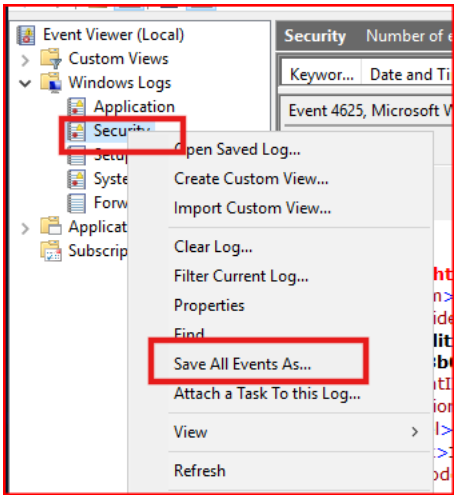
Security

Select:

Save All Events As...

Purpose:

- Creates evidence file
- Allows future investigation
- Similar to SOC incident handling workflow



In this lab, Windows Server 2019 auditing was successfully configured to capture security-related activities. Audit policies for **Logon Events, Account Logon Events, and Policy Changes** were enabled with **Success and Failure** tracking and verified using auditpol.

The network connectivity between all required VMs was confirmed, and the Windows Server was prepared as a security log source for future analysis.

Configuring, Viewing, and Analyzing IIS Logs

Configure **IIS (Internet Information Services) logging** on Windows Server 2019, generating web traffic, and analyzing IIS log entries to understand web server activity.

Objective

Verify IIS installation
Enable IIS logging
Configure required log fields
Generate HTTP requests
Locate IIS log files
Analyze web access logs

Machines Required

Machine	IP	Role
Windows Server 2019	192.168.100.130	IIS Web Server + Log Source
Attacker Machine	192.168.100.143	Generate web requests
Windows 10 Admin Machine	192.168.100.138	Log analysis

Tools Required

Tool	Purpose
IIS Manager	Configure web logging
Web Browser	Generate HTTP requests
Notepad / Text Editor	View IIS logs

No additional installation is required if IIS is already installed.

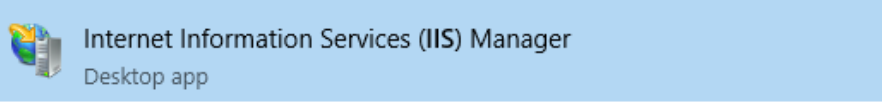
Configure IIS Logging

Enable and configure IIS logging on **Windows Server 2019** so that web requests are recorded for later analysis.

After this phase, IIS will capture:

- Client IP address
- Requested URL
- HTTP status codes
- User agent
- Referrer
- Bytes sent/received

Open IIS Manager



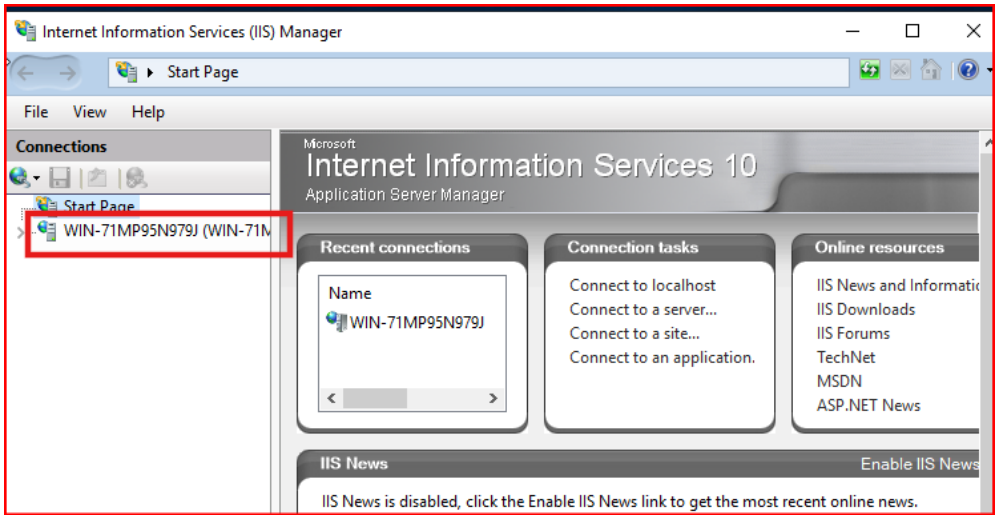
Select Website Logging

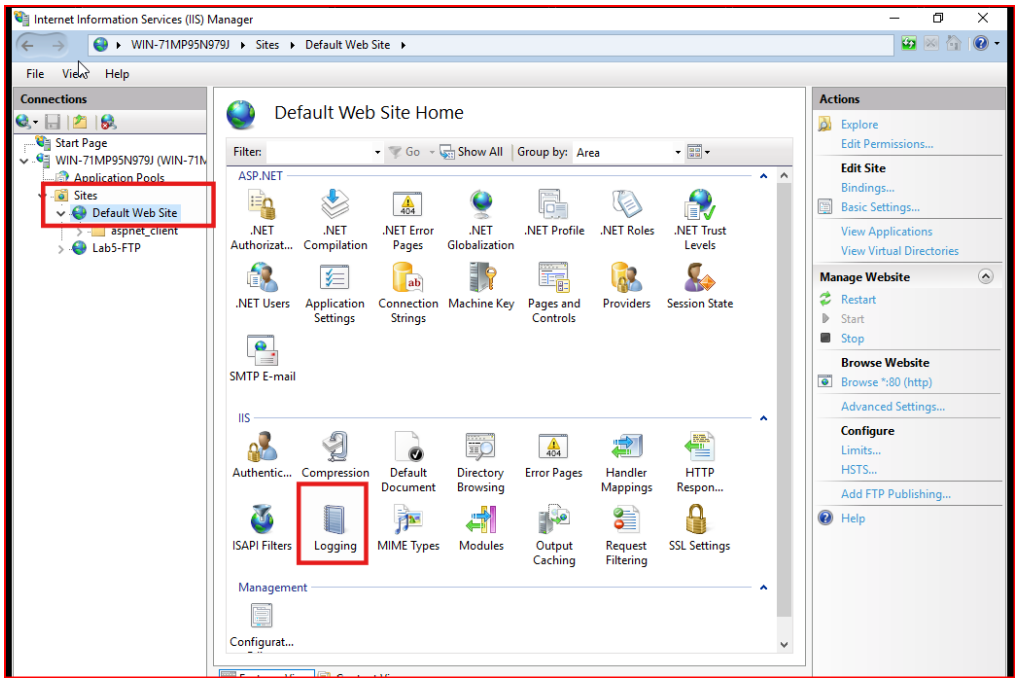
In IIS Manager:

Navigate: Connections-> Sites-> Default Web Site

Click: Default Web Site

Then open: Logging





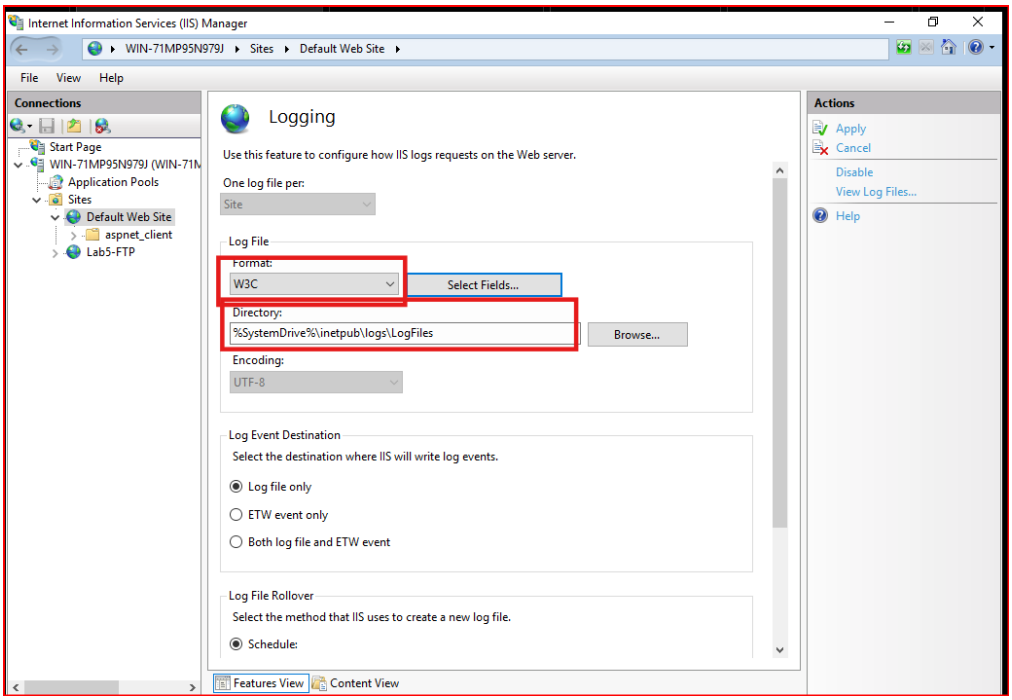
Enable Logging

In Logging settings:

Set: **Format** ->W3C

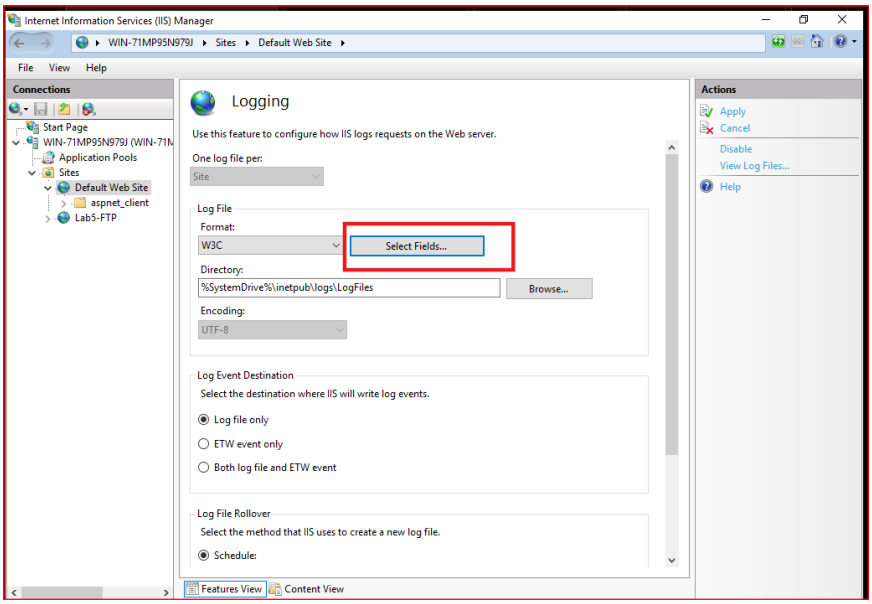
Directory ->Keep default:

C:\inetpub\logs\LogFiles

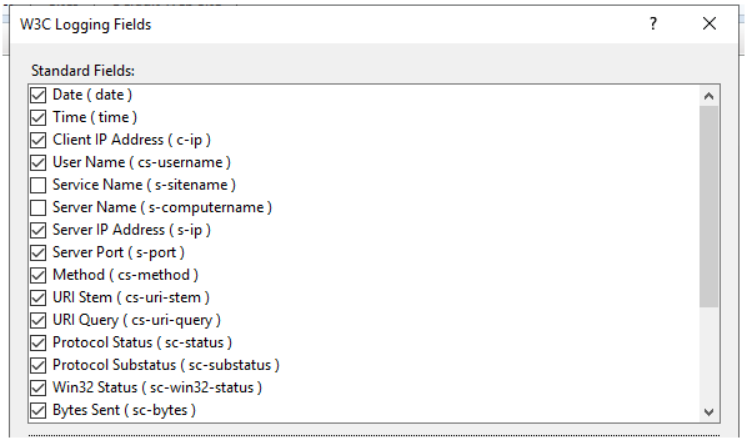


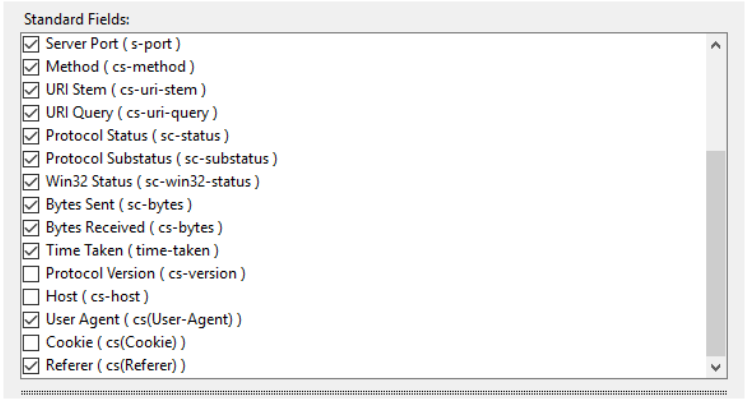
Configure Log Fields

Click: Select Fields...



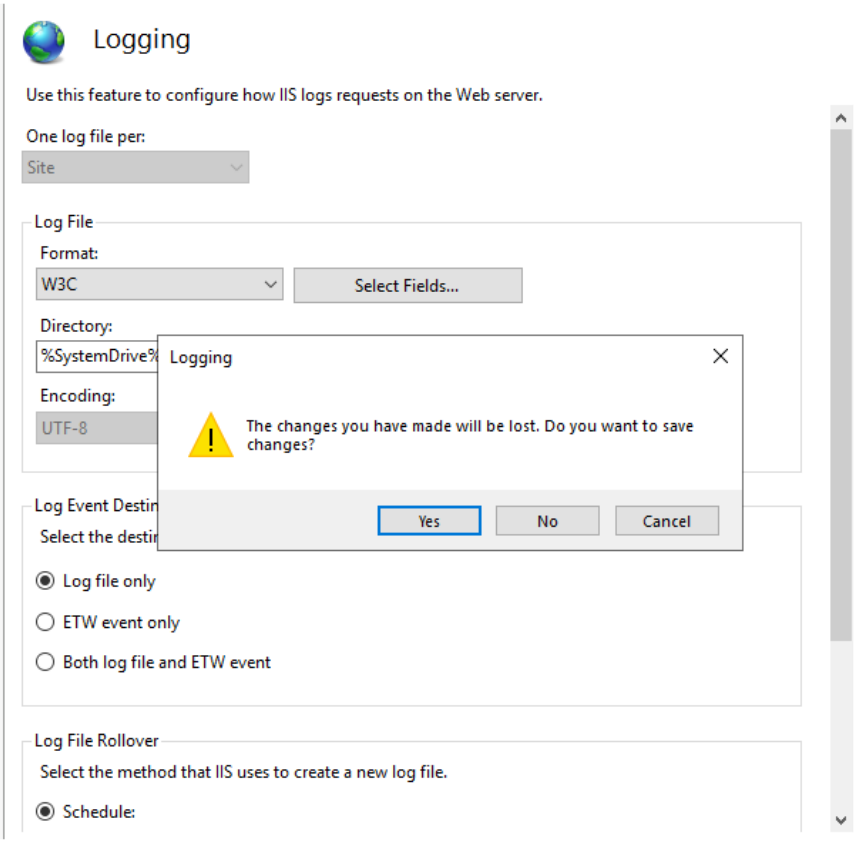
Select the fields as per selected below





Apply Configuration

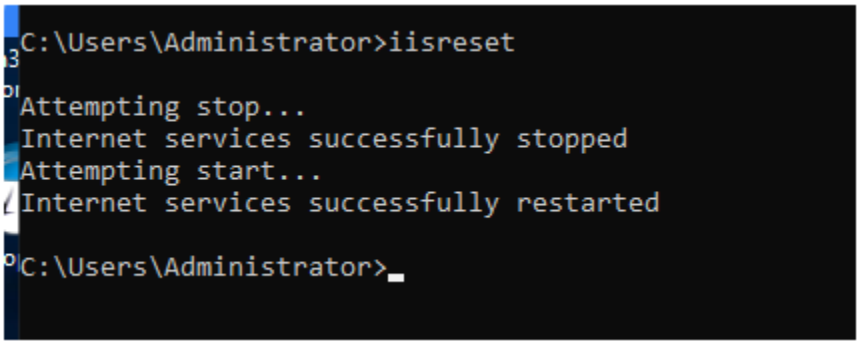
Click: Apply



Restart IIS Service

Open Administrator Command Prompt.

Run: iisreset

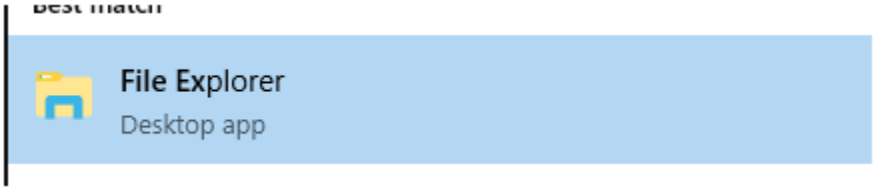


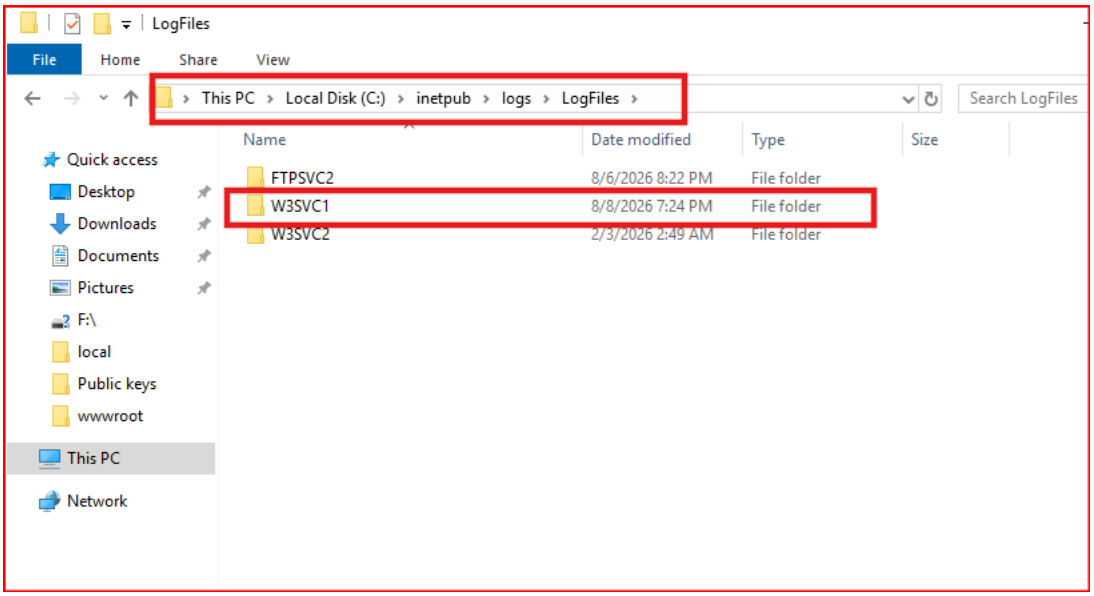
Verify Logging Directory

Open File Explorer:

Navigate:

C:\inetpub\logs\LogFiles

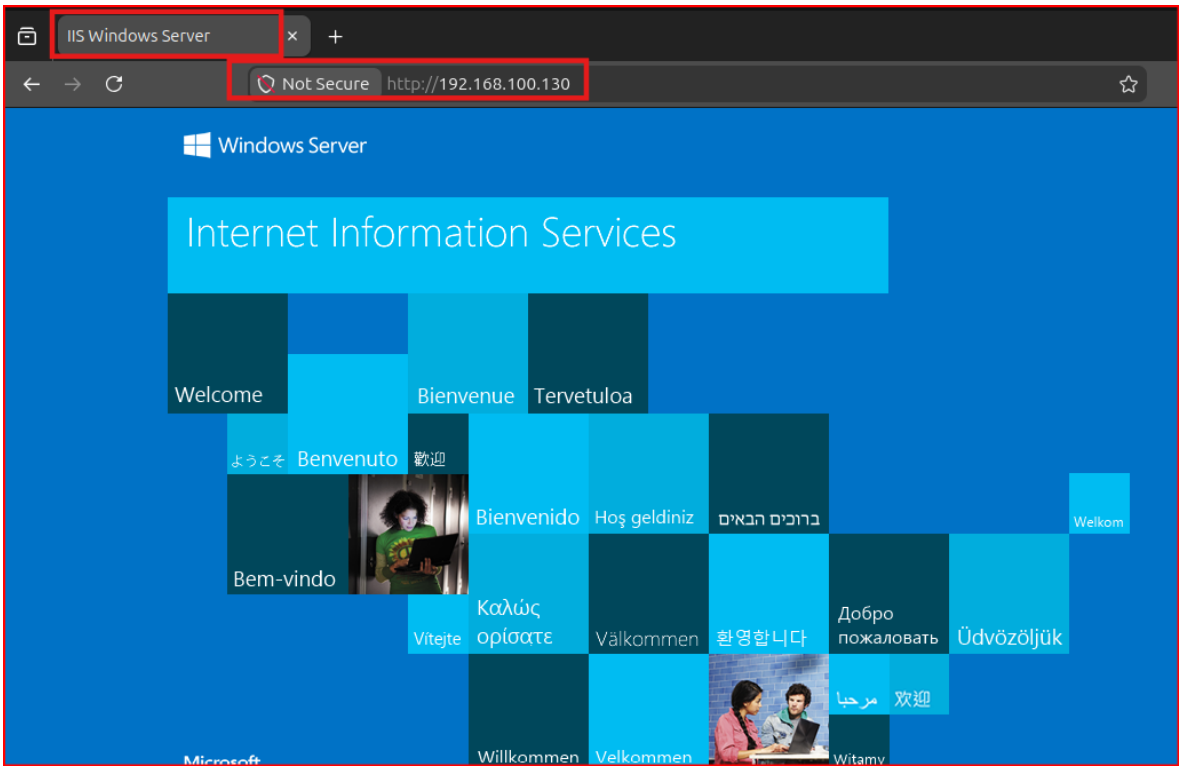




Verify IIS Website Access

Machine: Attacker Machine

Open browser. Enter: ->http://192.168.100.130



Generating Normal Web Requests

On Attacker Machine browser, open: http://192.168.100.130

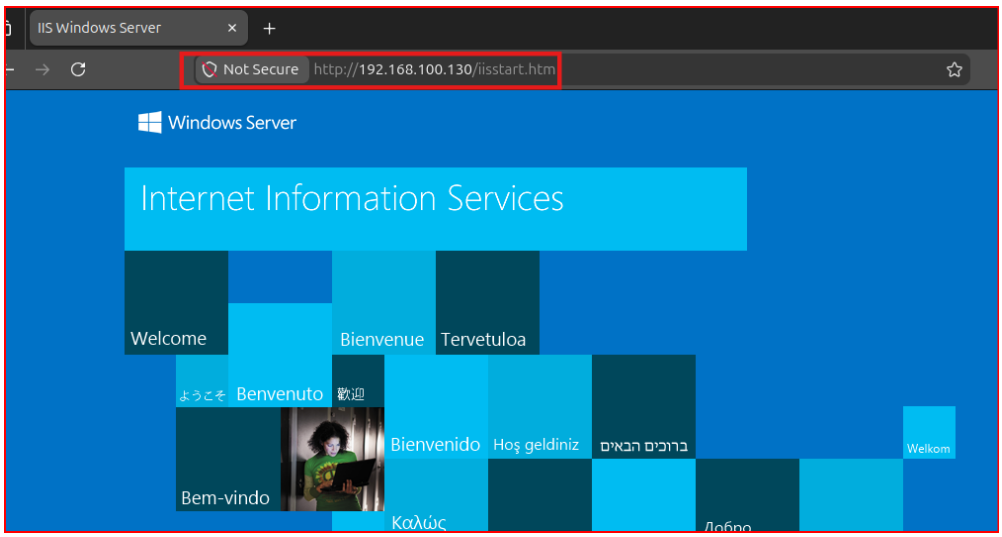
Refresh the page:

5-10 times

Then visit: http://192.168.100.130/iisstart.htm

Purpose:

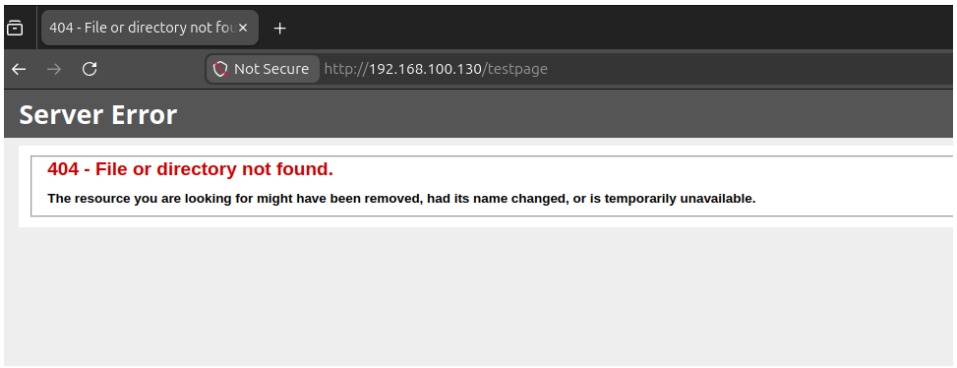
- Creates multiple HTTP GET requests
- Generates IIS log records



Generate Failed Requests

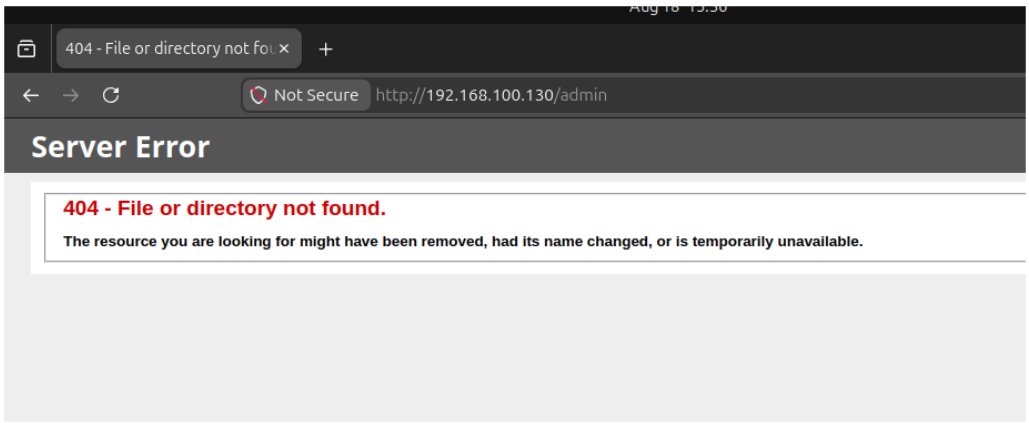
On Attacker Machine browser, open:

http://192.168.100.130/testpage



Then:

http://192.168.100.130/admin



Verify IIS Logs Are Updating

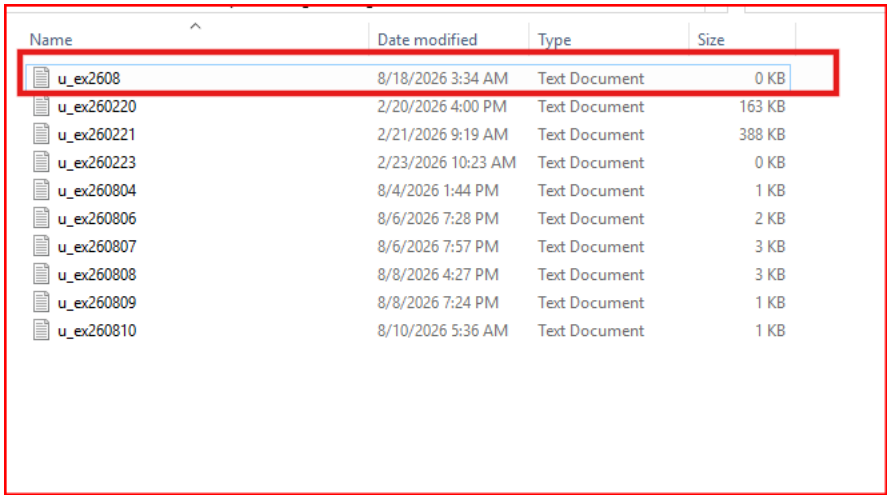
Machine: Windows Server 2019

Navigate: C:\inetpub\logs\LogFiles\W3SVC1

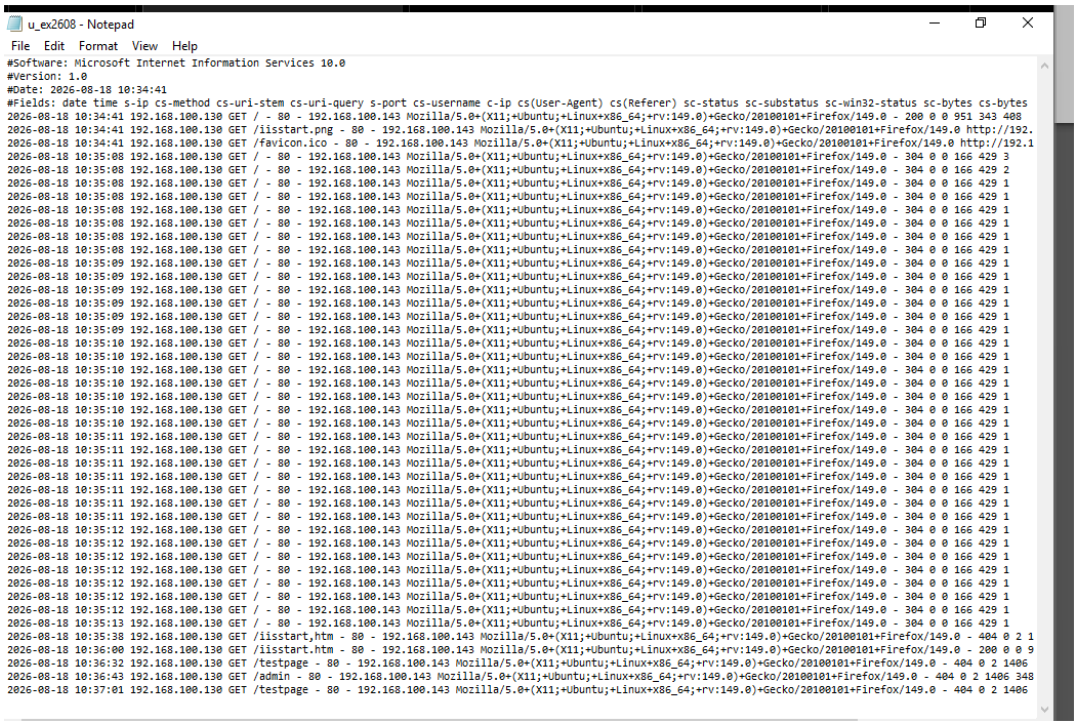
Open the latest:

Example: u_ex260806.log

Expected entries:



Name	Date modified	Type	Size
u_ex2608	8/18/2026 3:34 AM	Text Document	0 KB
u_ex260220	2/20/2026 4:00 PM	Text Document	163 KB
u_ex260221	2/21/2026 9:19 AM	Text Document	388 KB
u_ex260223	2/23/2026 10:23 AM	Text Document	0 KB
u_ex260804	8/4/2026 1:44 PM	Text Document	1 KB
u_ex260806	8/6/2026 7:28 PM	Text Document	2 KB
u_ex260807	8/6/2026 7:57 PM	Text Document	3 KB
u_ex260808	8/8/2026 4:27 PM	Text Document	3 KB
u_ex260809	8/8/2026 7:24 PM	Text Document	1 KB
u_ex260810	8/10/2026 5:36 AM	Text Document	1 KB



u_ex2608 - Notepad

File Edit Format View Help

```
#Software: Microsoft Internet Information Services 10.0
#Version: 1.0
#Date: 2026-08-18 10:34:41
#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status sc-bytes cs-bytes
2026-08-18 10:34:41 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 200 0 0 951 343 408
2026-08-18 10:34:41 192.168.100.130 GET /iisstart.png - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 http://192.
2026-08-18 10:34:41 192.168.100.130 GET /favicon.ico - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 http://192.1
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 3
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 2
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:08 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:09 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:09 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:09 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:09 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:09 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:09 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:10 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:10 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:10 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:10 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:10 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:10 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:11 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:11 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:11 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:11 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:11 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:11 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:12 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:12 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:12 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:12 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:12 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:12 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:13 192.168.100.130 GET / - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 304 0 0 166 429 1
2026-08-18 10:35:38 192.168.100.130 GET /iisstart.htm - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 404 0 2 1
2026-08-18 10:36:00 192.168.100.130 GET /iisstart.htm - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 200 0 0 9
2026-08-18 10:36:32 192.168.100.130 GET /testpage - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 404 0 2 1406
2026-08-18 10:36:43 192.168.100.130 GET /admin - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 404 0 2 1406 348
2026-08-18 10:37:01 192.168.100.130 GET /testpage - 80 - 192.168.100.143 Mozilla/5.0+(X11;+Ubuntu;+Linux+x86_64;+rv:149.0)+Gecko/20100101+Firefox/149.0 - 404 0 2 1406
```

In this lab, IIS logging was successfully configured on **Windows Server 2019** to capture web server activity. Web traffic was generated from the attacker machine, and IIS log files were verified in the IIS log directory.

The lab demonstrated how IIS records important web activity information, including:

- Client IP address
- HTTP requests
- Requested resources
- Response status codes
- Web access activity

The IIS logs were successfully generated and verified, confirming that the web server is properly collecting access information.

Centralized Log Management Using Splunk

Install and configure **Splunk Enterprise** and **Splunk Universal Forwarder** to collect Windows and IIS logs centrally for monitoring and analysis.

Objective

Install Splunk Enterprise on Admin Machine
Configure Splunk receiving service
Install Splunk Universal Forwarder on Web Server
Forward Windows Event Logs
Forward IIS Logs
Search and analyze collected logs in Splunk

Machines Required

Machine	IP	Role
Windows 10 Admin Machine	192.168.100.138	Splunk Enterprise Server
Windows Server 2019	192.168.100.130	Log Source + Universal Forwarder
Attacker Machine	192.168.100.143	Generate activity (if required)

Tools Required

Tool	Purpose
Splunk Enterprise	Centralized log analysis platform
Splunk Universal Forwarder	Sends logs to Splunk server
Windows Event Logs	Security log source
IIS Logs	Web activity log source

Machine: Windows 10 Admin Machine

SPLUNK ENTERPRISE INSTALLATION & SETUP

Open **PowerShell as Administrator**:

```
wget -O splunk-enterprise.msi
https://download.splunk.com/products/splunk/releases/10.4.2/windows/splunk-10.4.2-33c3bf42cd73-windows-x64.msi
```

```
PS C:\Windows\system32> wget -O splunk-enterprise.msi "https://download.splunk.com/products/splunk/releases/10.4.2/windows/splunk-10.4.2-33c3bf42cd73-windows-x64.msi"
```

```
Last 200 Keys:
 6 4 . m s i " Enter
 Enter
 d i r Space s p l u n k - e n t e r p r i s e . m s i Enter
 m s i . e x e Space / i Space s p l u n k - e n t e r p r i s e , m s i Enter
 UpArrow LeftArrow LeftArrow LeftArrow Backspace . Enter
 DownArrow UpArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow L
eftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow L
eftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow e x e c Enter
 UpArrow UpArrow UpArrow UpArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArr
ow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArrow LeftArr
ow LeftArrow LeftArrow LeftArrow LeftArrow RightArrow RightArrow RightArrow Backspace Backspace e l Enter
 UpArrow UpArrow UpArrow UpArrow DownArrow DownArrow DownArrow DownArrow DownArrow DownArrow DownArrow DownArrow DownArr
ow DownArrow DownArrow DownArrow DownArrow DownArrow DownArrow DownArrow DownArrow c l e a r Enter
 UpArrow UpArrow UpArrow UpArrow UpArrow UpArrow UpArrow Enter
 UpArrow UpArrow UpArrow UpArrow RightArrow RightArrow RightArrow RightArrow RightArrow RightArrow RightArrow RightArrow
 RightArrow RightArrow PageDown

Exception:
System.IO.IOException: The parameter is incorrect.

   at System.IO.__Error.WinIOError(Int32 errorCode, String maybeFullPath)
   at System.Console.SetWindowPosition(Int32 left, Int32 top)
   at Microsoft.PowerShell.PSConsoleReadLine.ScrollDisplayDown(Nullable`1 key, Object arg)
   at Microsoft.PowerShell.PSConsoleReadLine.ProcessOneKey(ConsoleKeyInfo key, Dictionary`2 dispatchTable, Boolean ignor
eIfNoAction, Object arg)
   at Microsoft.PowerShell.PSConsoleReadLine.InputLoop()
   at Microsoft.PowerShell.PSConsoleReadLine.ReadLine(Runspace runspace, EngineIntrinsics engineIntrinsics)
```

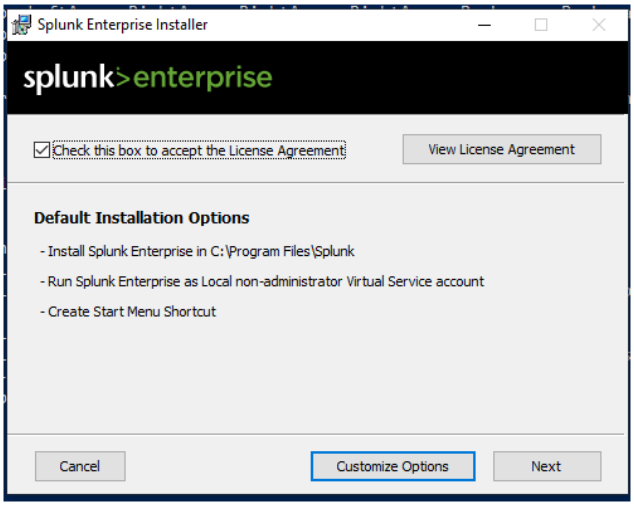
Splunk provides the Windows MSI package for Enterprise installation.

Installation:

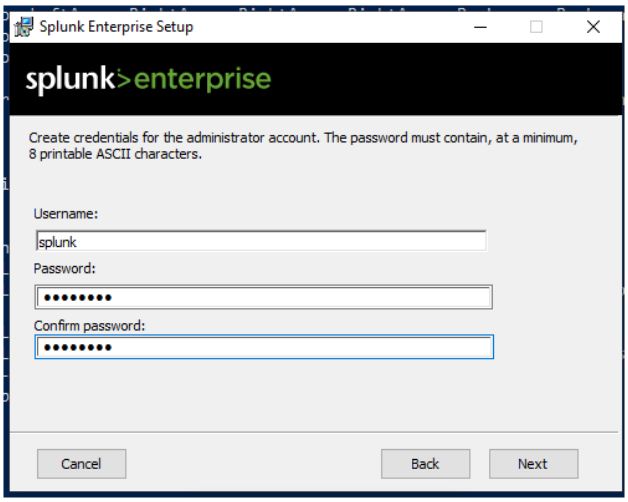
```
msiexec.exe /i splunk-enterprise.msi
```

```
PS C:\Windows\system32> msiexec.exe /i splunk-enterprise.msi
PS C:\Windows\system32>
```

Accept license & Click Next



Add Credentials



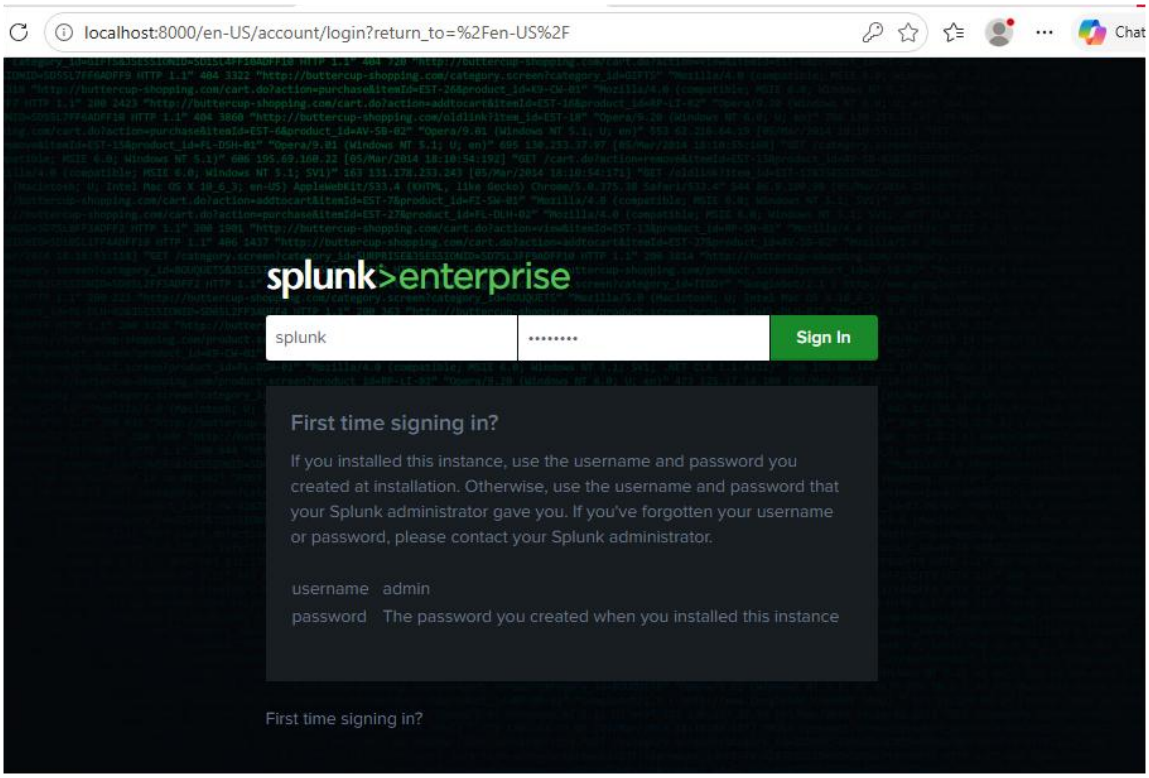
Waiting for Installation

Verify Splunk Home Dashboard

Sign in with the Credentials we set before

Username: splunk

Password: Admin123

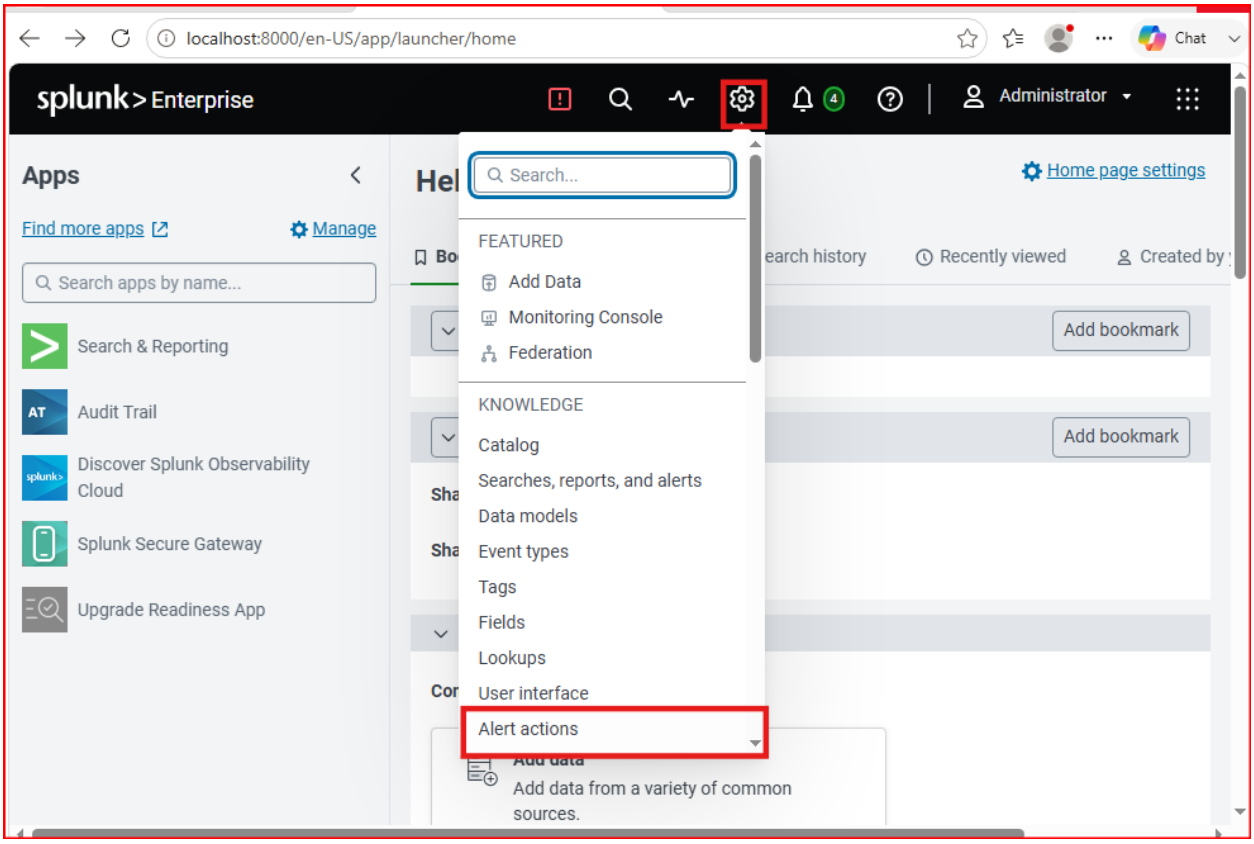


VERIFY RECEIVING PORT

You're logged in successfully. Now verify port 9997 is enabled.

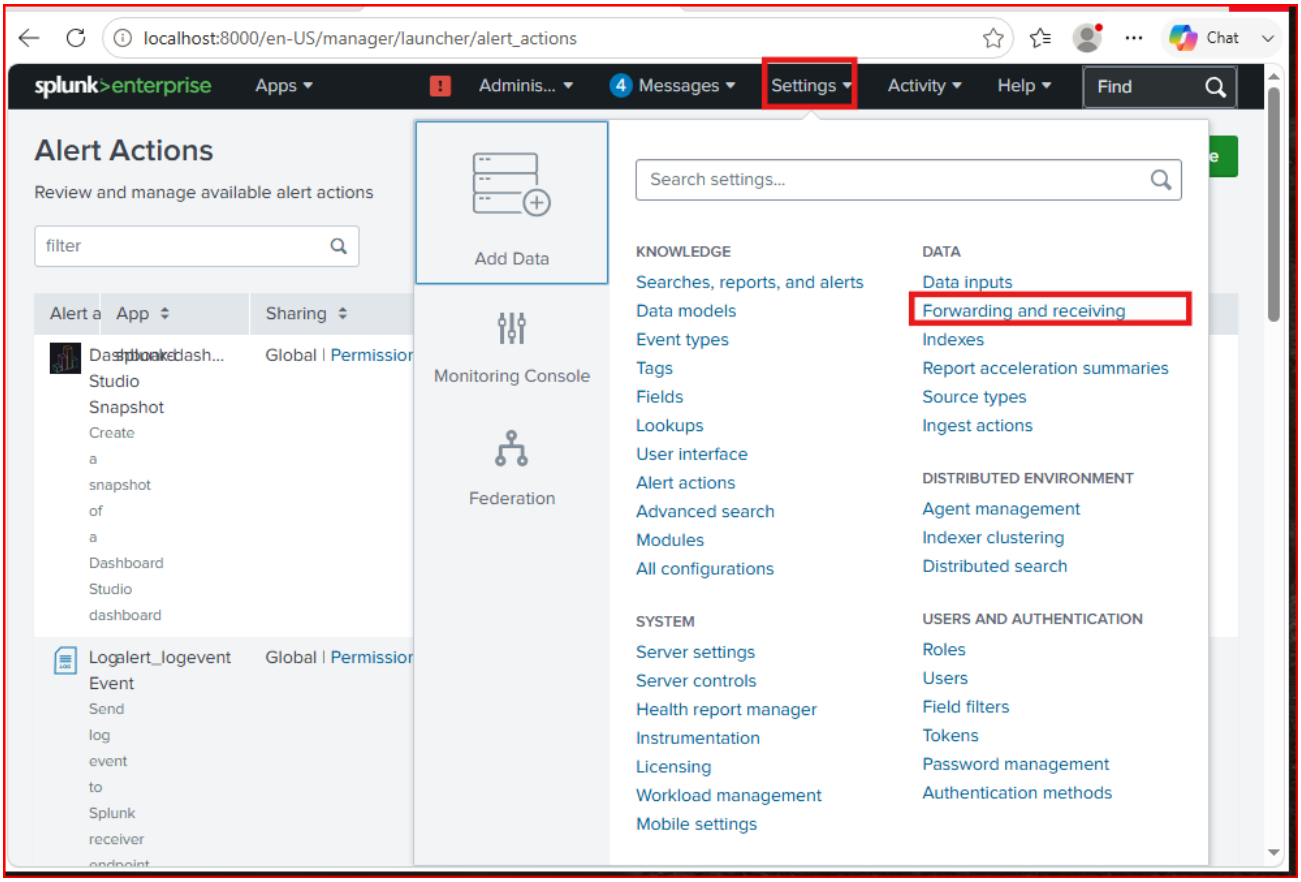
Go to Settings

1. Click **Settings** (gear icon, top right)
2. Look for dropdown menu
3. Select Alert actions



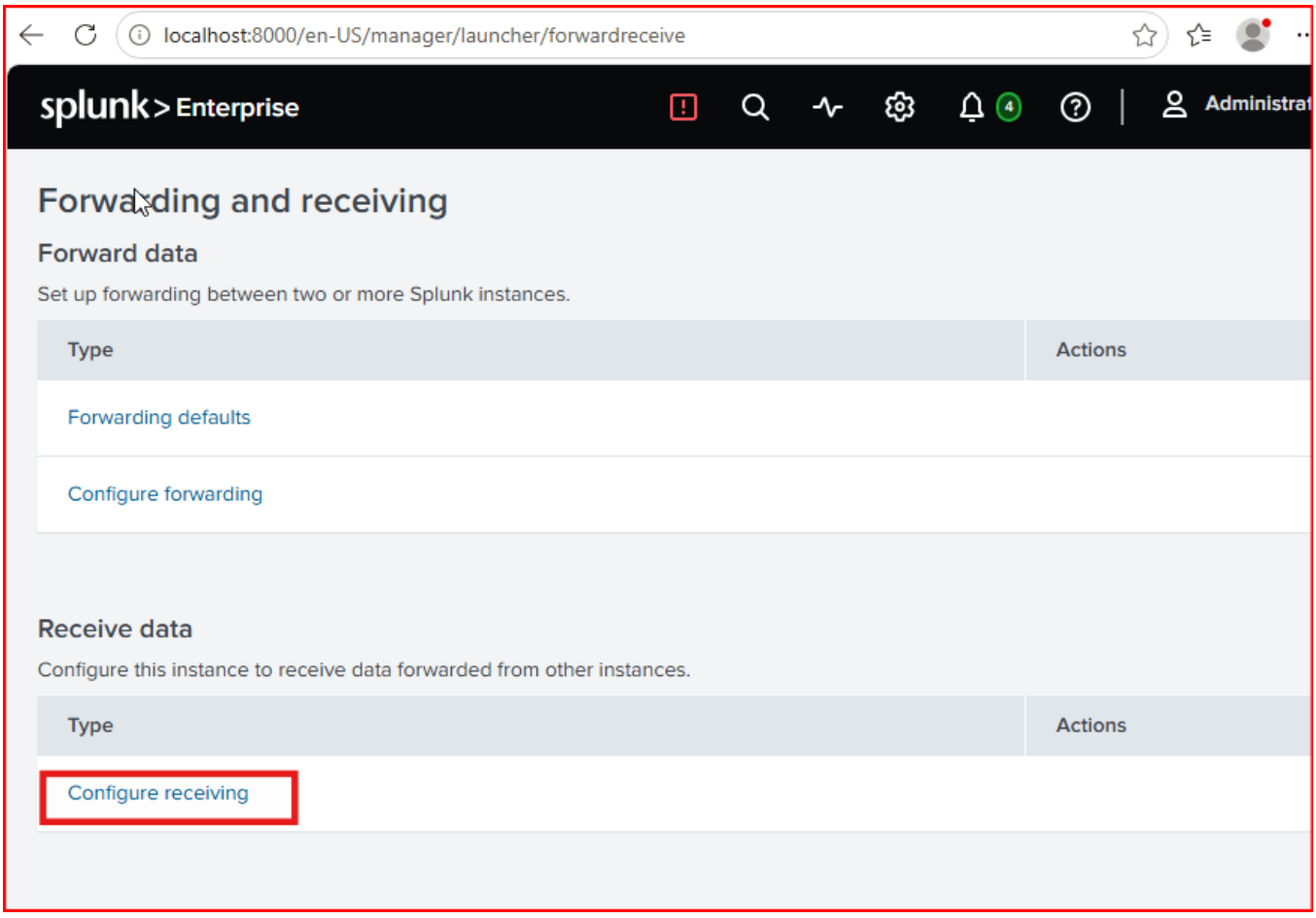
Find Forwarding and Receiving

1. Click **Forwarding and receiving**
2. You should see options:
    - Configure receiving
    - Configure forwarding



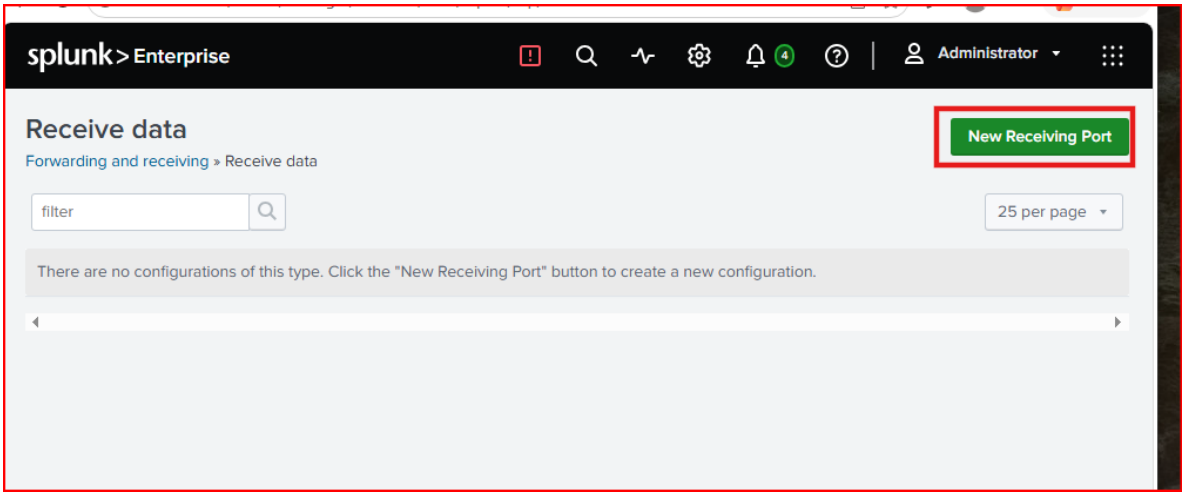
Configure Receiving

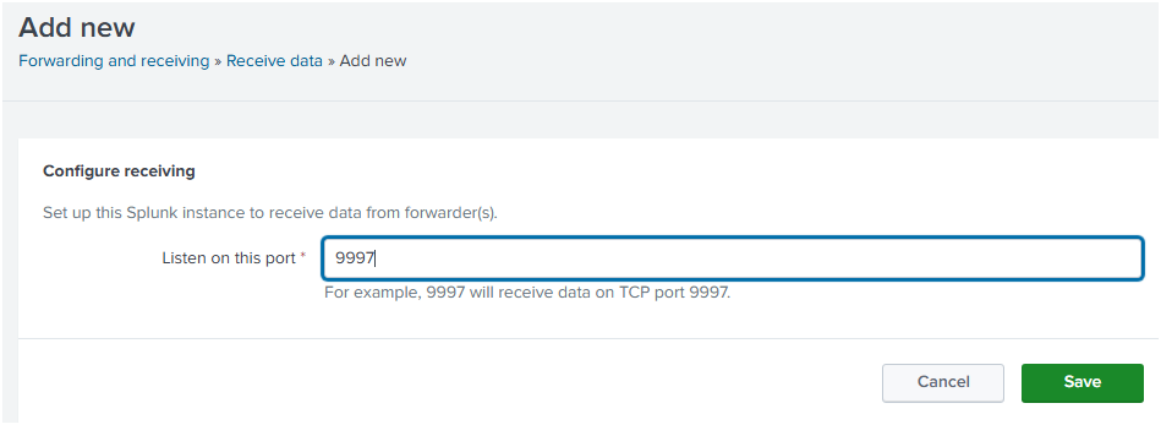
1. Click **Configure receiving**
2. Check if port **9997** is listed

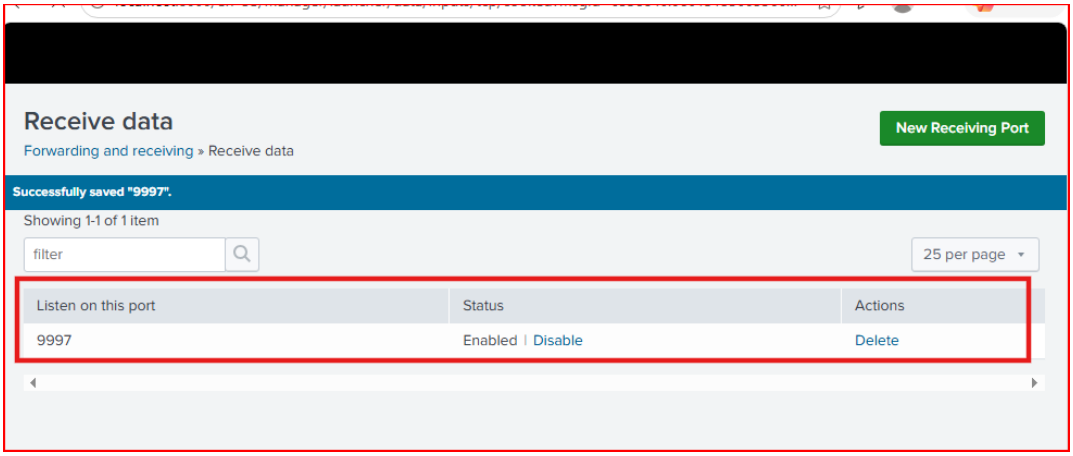


If port 9997 is NOT listed → Do this:

1. Click **New Receiving Port**
2. Enter: 9997
3. Click **Save**







Machine: Windows Server 2019 (Web Server)

CONFIGURE SPLUNK RECEIVER

Test Connection from Web Server

Open **PowerShell** and run:

```
Test-NetConnection -ComputerName 192.168.100.138 -Port 9997
```

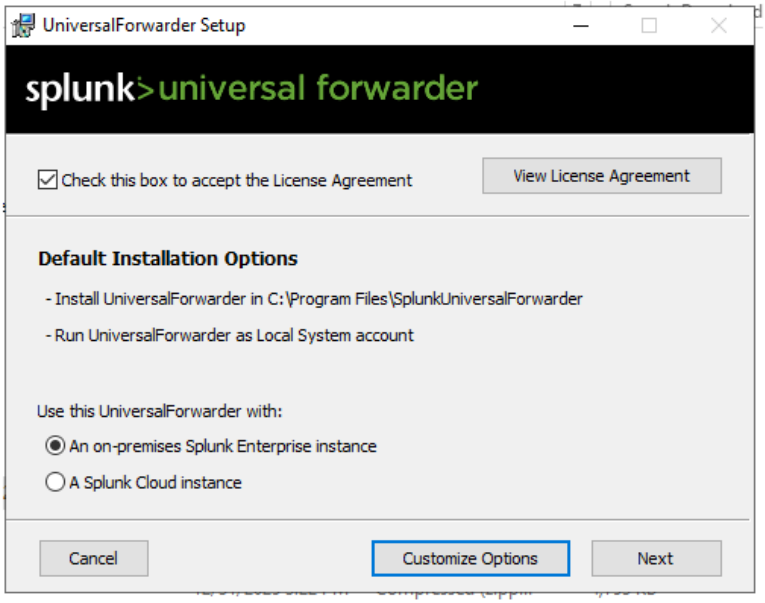
```
PS C:\Users\Administrator> Test-NetConnection -ComputerName 192.168.100.138 -Port 9997


ComputerName     : 192.168.100.138
RemoteAddress    : 192.168.100.138
RemotePort       : 9997
InterfaceAlias   : Ethernet0 2
SourceAddress    : 192.168.100.130
TcpTestSucceeded : True
```

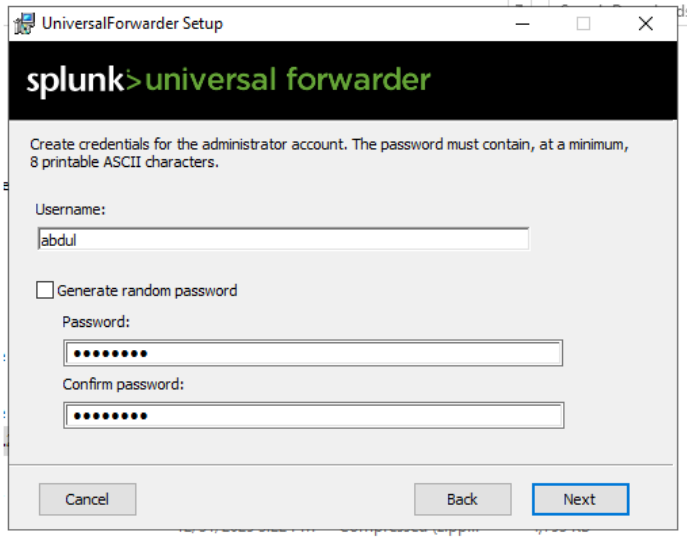
INSTALL SPLUNK UNIVERSAL FORWARDER

installs Splunk Universal Forwarder on Windows Server 2019 (192.168.100.130). The forwarder collects logs from Windows Event Logs and IIS, then sends them to Splunk Enterprise on the Admin Machine.

- **Complete Installation Wizard**
- **Accept License Agreement**
- **Verify Default Options**
- **Click Next**



Set ADMIN CREDENTIALS

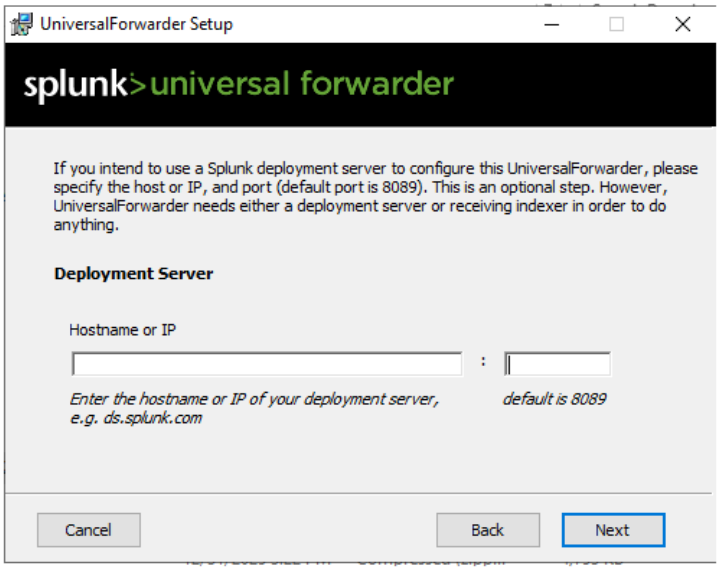


DEPLOYMENT SERVER (RECEIVING INDEXER)

- **Deployment Server** = Optional (centrally manages configurations)
- **Receiving Indexer** = Required (where logs are sent) ← **THIS ONE**

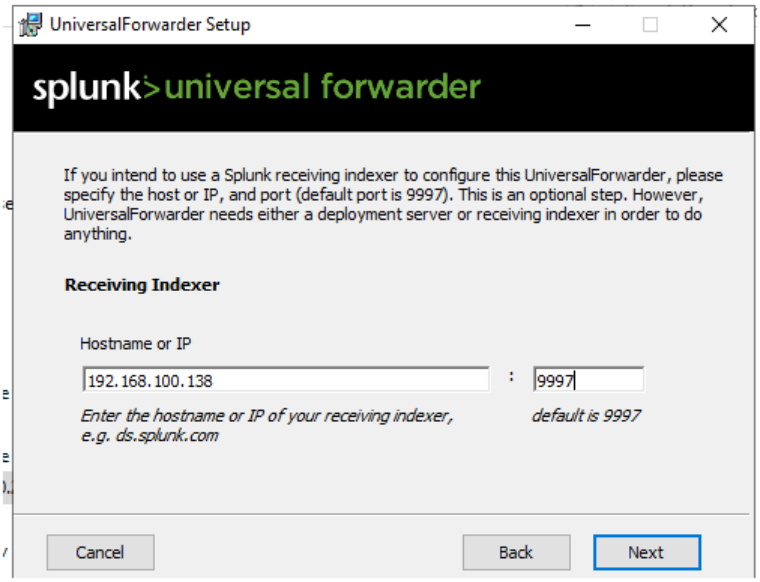
For this lab, we ONLY need **Receiving Indexer**.

Click Next (default empty)



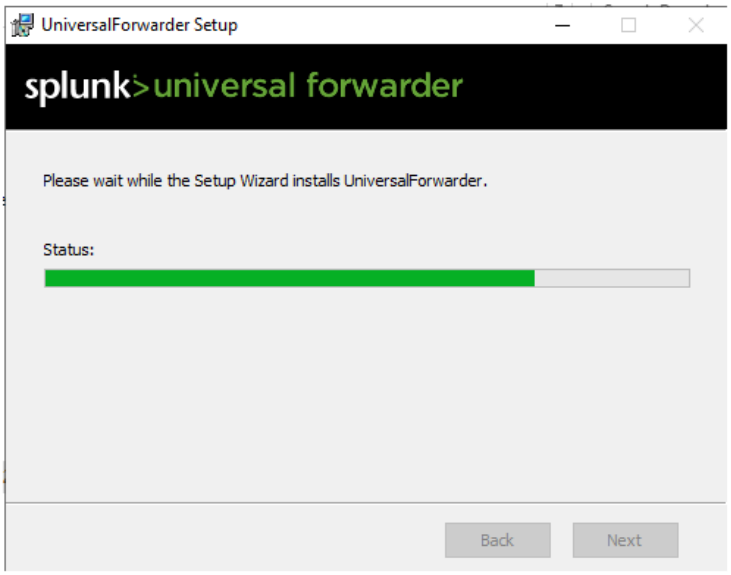
Enter Receiving Indexer

1. Hostname or IP field → Type: 192.168.100.138
2. Port field → Verify: 9997 (default - keep it)



Waiting for Installation

- Estimated time: 3-5 minutes
- Do NOT close window

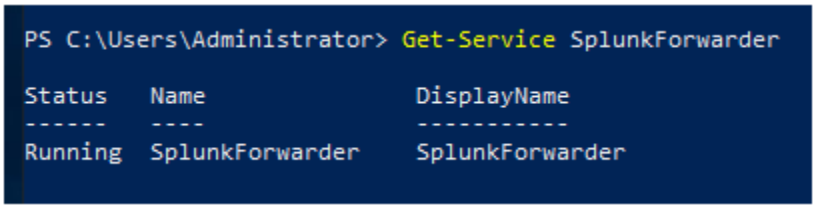


Verify Installation

After installation completes:

```
Get-Service SplunkForwarder
```

Expected Output:



CONFIGURE LOG FORWARDING

Configures what logs Universal Forwarder should collect and send to Splunk Enterprise. We need to configure it forward:

1. **Windows Event Logs** (Security, System, Application)
2. **IIS Logs** (from C:\inetpub\logs\LogFiles)

Locate Forwarder Configuration File

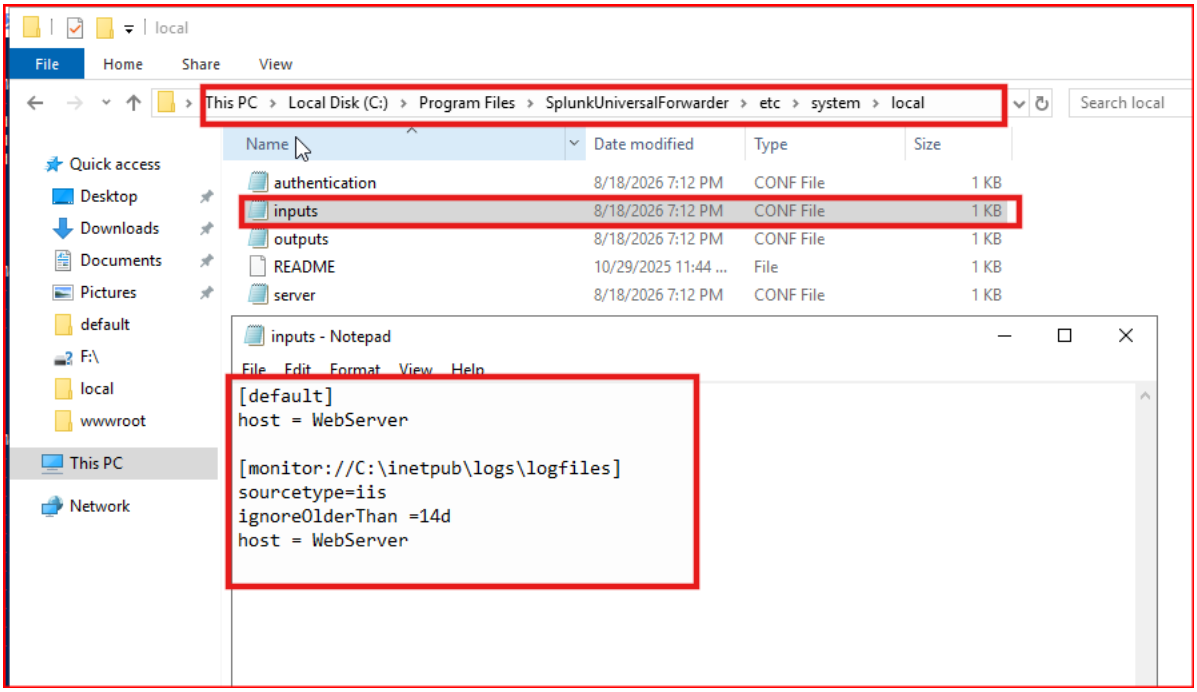
On **Windows Server 2019** (192.168.100.130), open **File Explorer**:

C:\Program Files\SplunkUniversalForwarder\etc\system\local

Find file: inputs.conf (if not present create one)

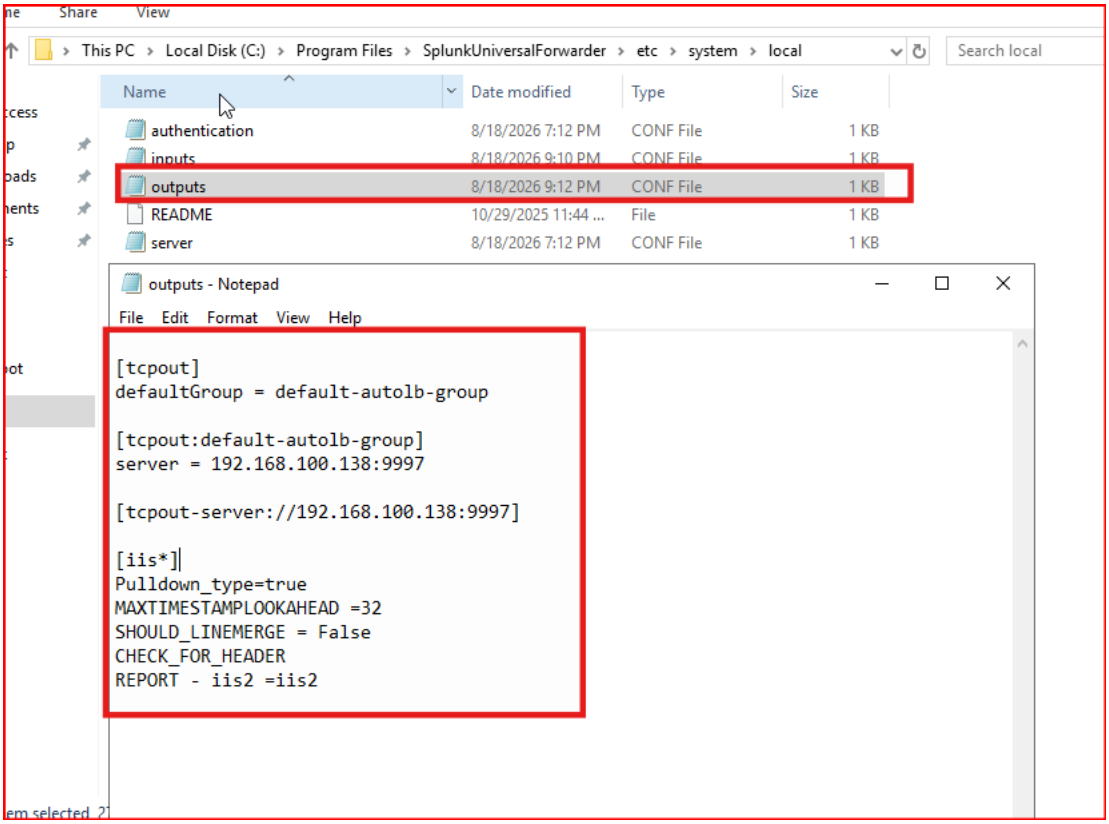
Edit inputs.conf

1. **Open with Notepad++** (or Notepad)
2. Clear all content
3. Paste this configuration:

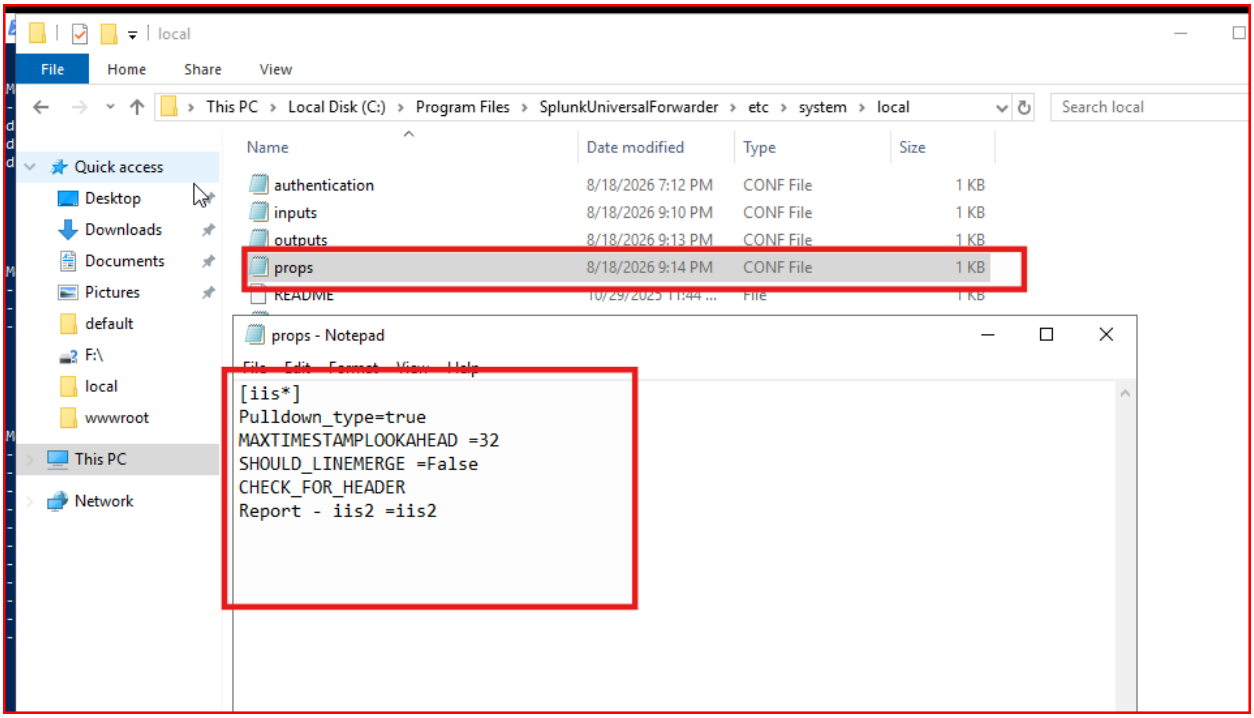


Save the file

Edit Output.conf File

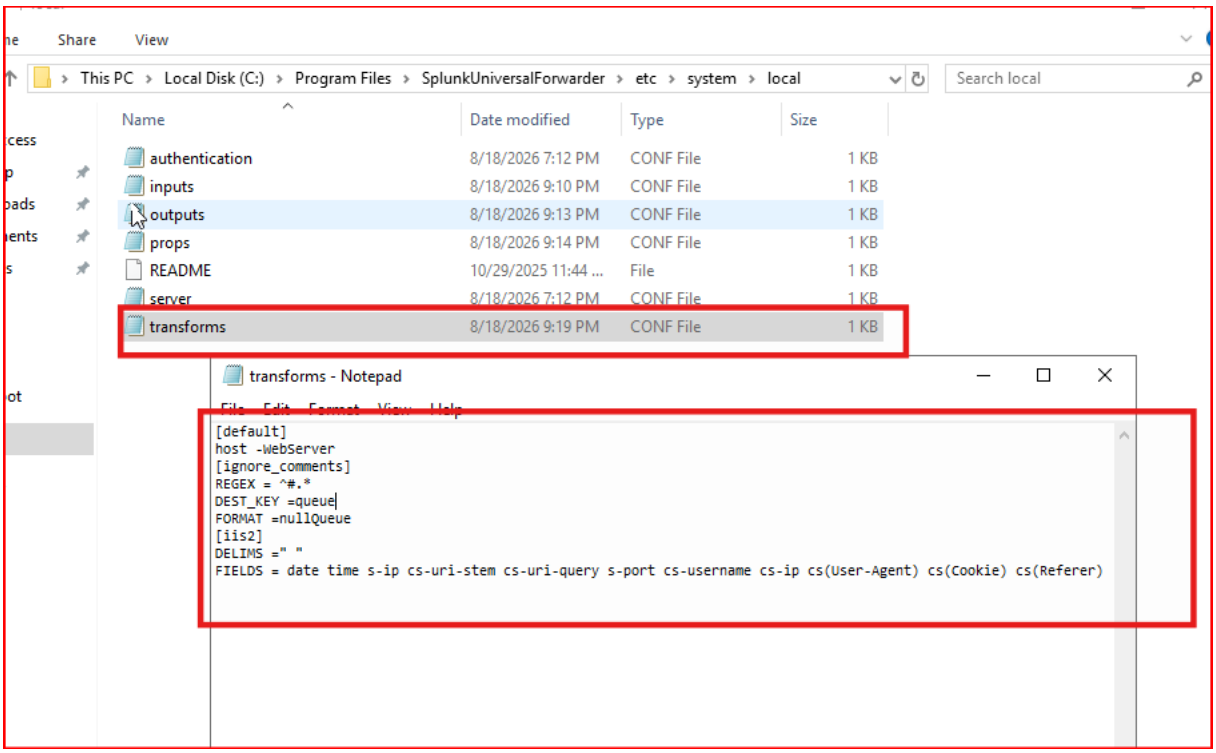


Edit prpos.conf file (create if not exist)



Edit transforms.conf file

transforms.conf file



Restart SplunkForwarder

```
PS C:\Users\Administrator> Restart-Service SplunkForwarder
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
PS C:\Users\Administrator>
```

Verify Service is Running

```
Get-Service SplunkForwarder
```

```
PS C:\Users\Administrator> Get-Service SplunkForwarder

Status   Name               DisplayName
------   ----               -----------
Running  SplunkForwarder    SplunkForwarder
```

Verify Forwarder is Connected to Splunk

```
Get-EventLog -LogName System -Newest 10 | findstr "SplunkForwarder"
```

```
PS C:\Users\Administrator> Get-EventLog -LogName System -Newest 10 | findstr "SplunkForwarder"
PS C:\Users\Administrator>
```

VERIFY CENTRALIZED LOGGING

Verifies that Splunk Enterprise is receiving logs from the Universal Forwarder on the Web Server. We check the Splunk Web interface to confirm logs are flowing in from both Windows Event Logs and IIS.

Go Back to Splunk Web

On **Windows 10 Admin Machine** (192.168.100.138):

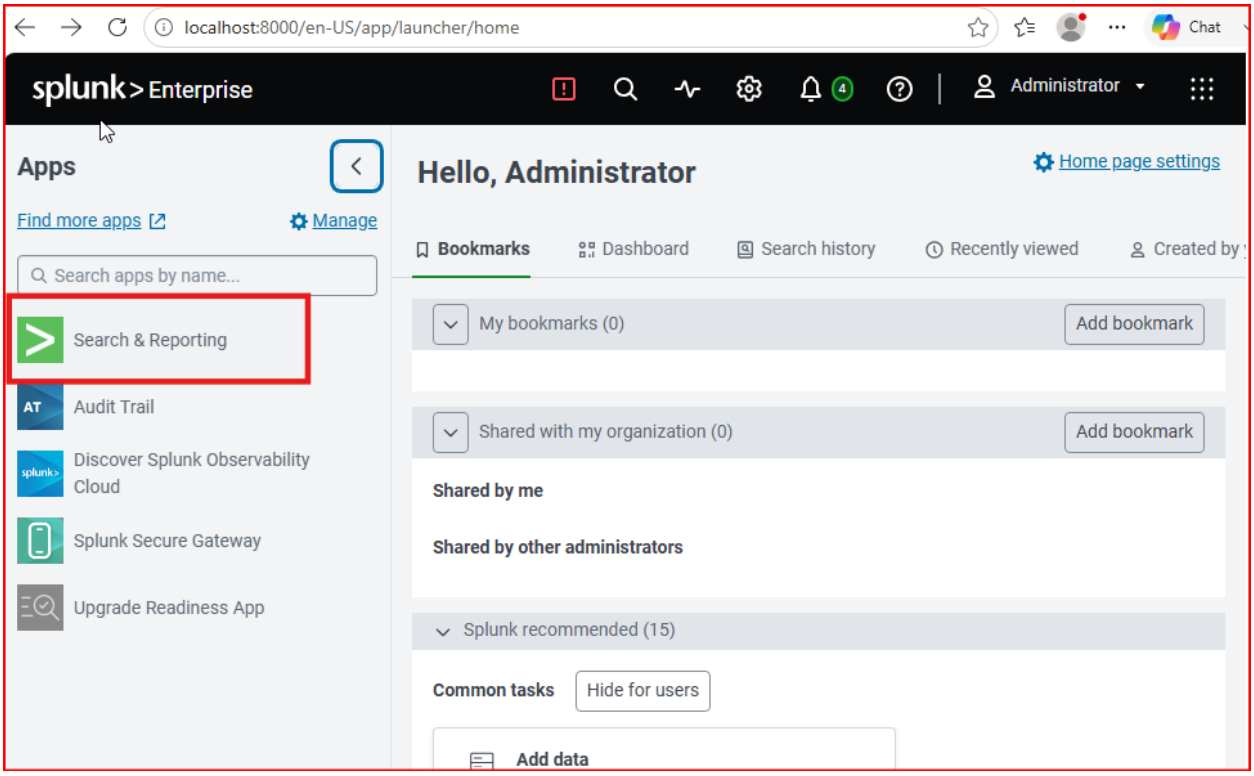
Open browser → http://localhost:8000 (or http://192.168.100.138:8000)

Login if needed:

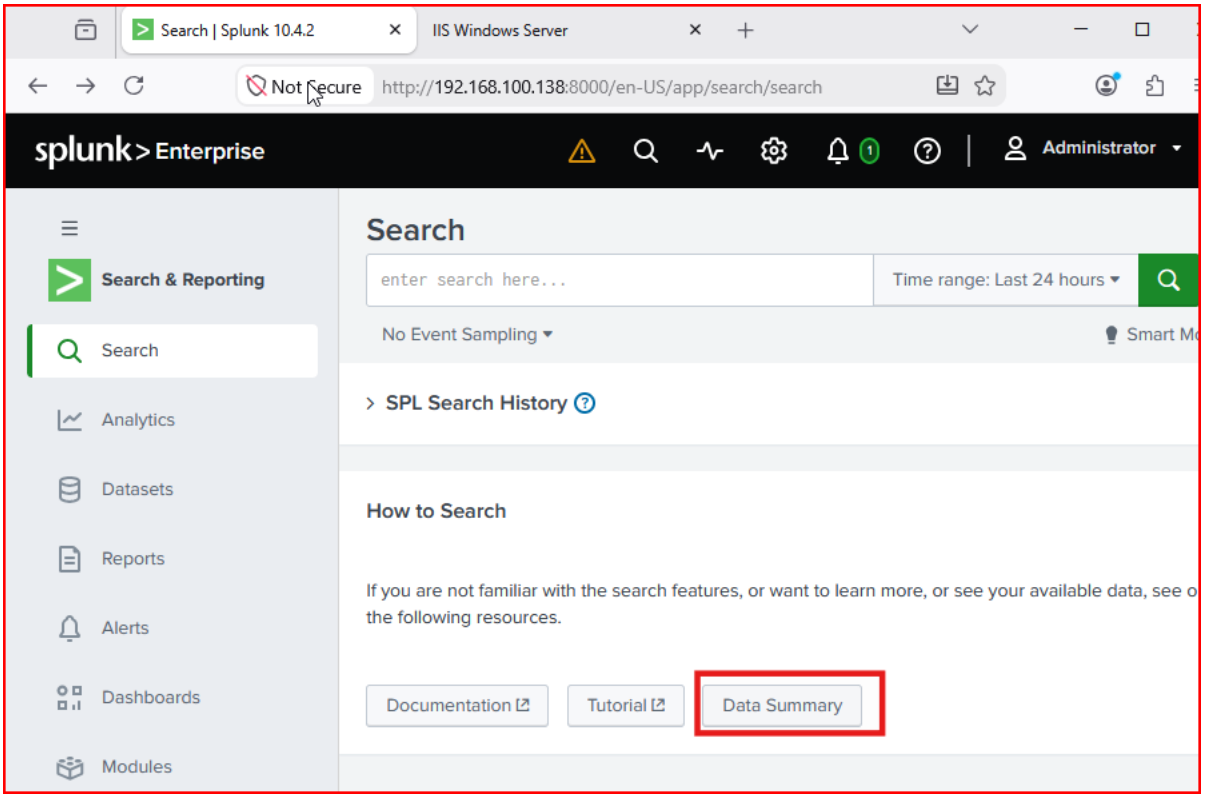
- Username: admin
- Password: Splunk@2024

Search for Windows Event Logs

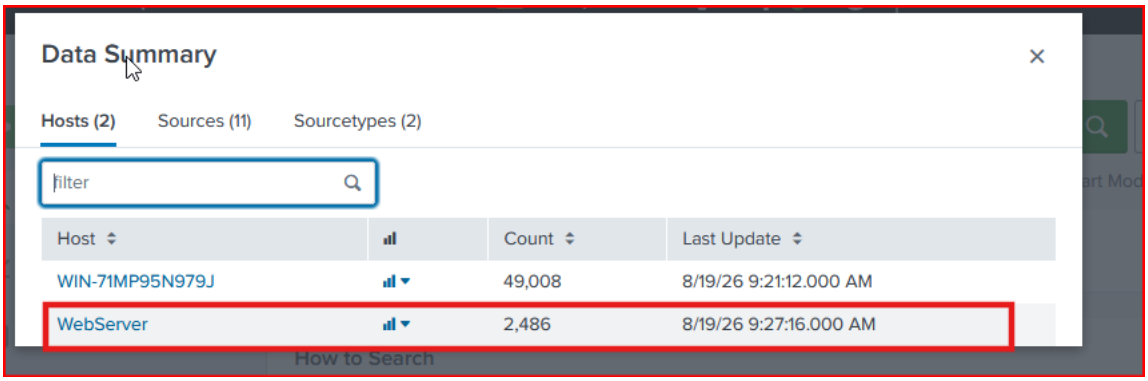
1. Click **Search & Reporting** (left sidebar)

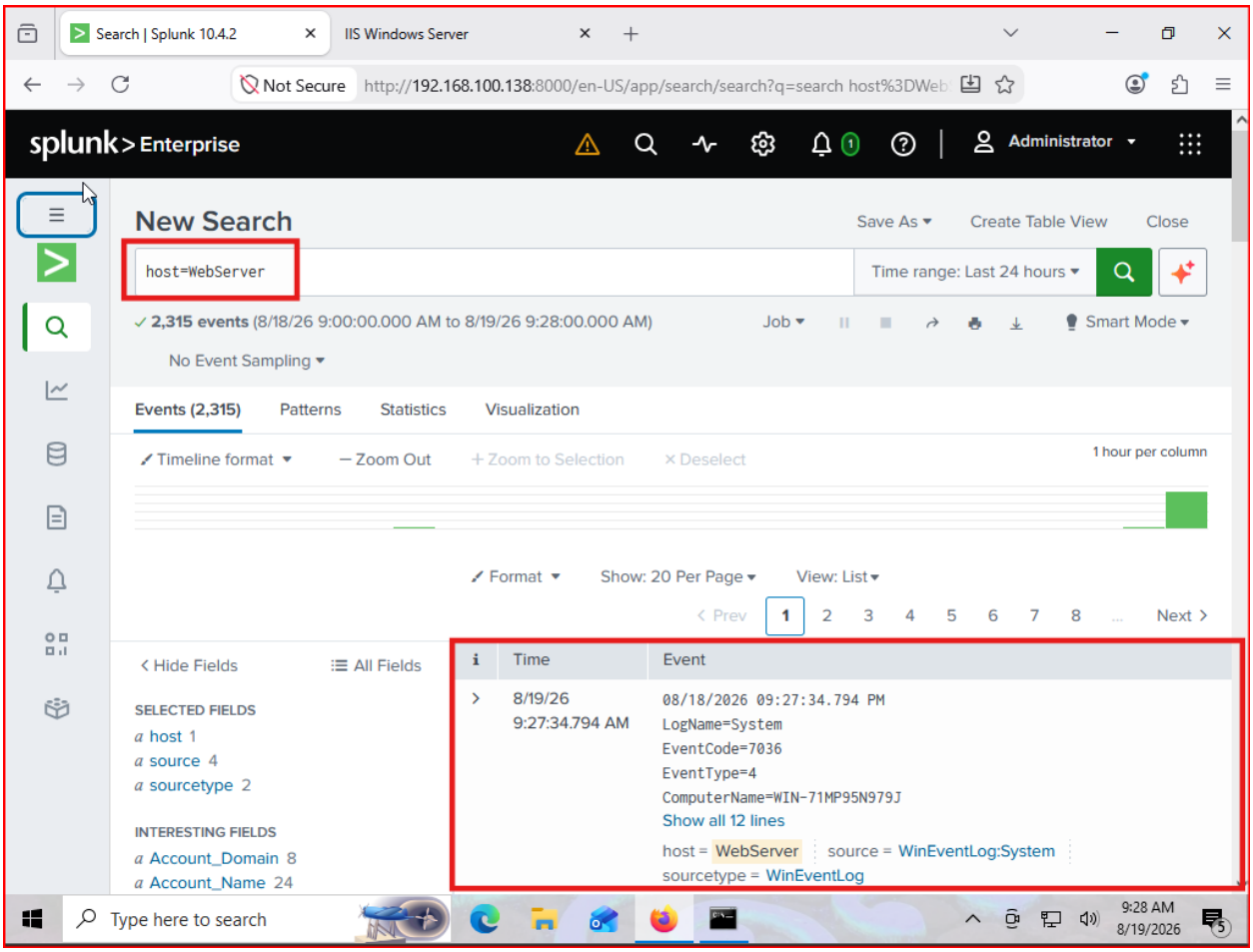


The Search console appears; **click on Data Summary** under the How to Search section



The host list will appear. Select the host machine (here, WebServer)



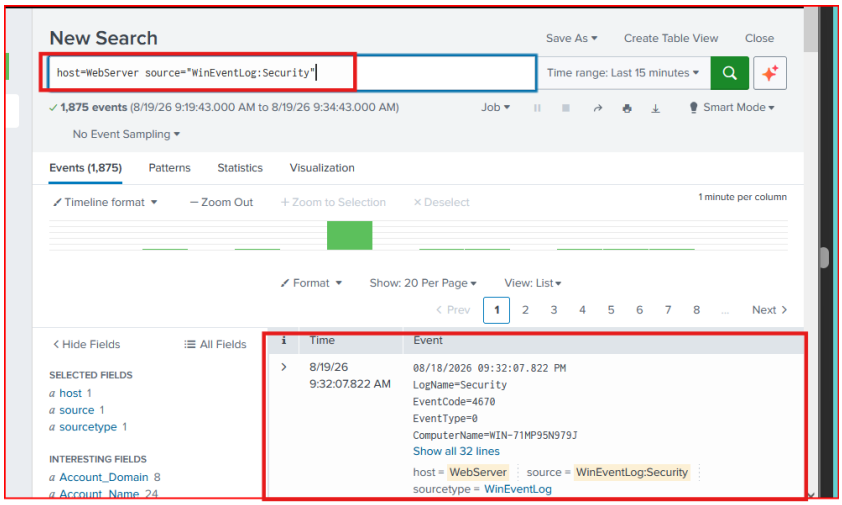

Search | Splunk 10.4.2
IIS Windows Server
Not Secure
http://192.168.100.138:8000/en-US/app/search/search?q=search host%3DWeb
splunk> Enterprise
Administrator
New Search
Save As
Create Table View
Close
host=WebServer
Time range: Last 24 hours
2,315 events (8/18/26 9:00:00.000 AM to 8/19/26 9:28:00.000 AM)
Job
Smart Mode
No Event Sampling
Events (2,315)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 hour per column
Format
Show: 20 Per Page
View: List
Prev
1 2 3 4 5 6 7 8 ...
Next
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 4
a sourcetype 2
INTERESTING FIELDS
a Account_Domain 8
a Account_Name 24
Time
Event
8/19/26
9:27:34.794 AM
08/18/2026 09:27:34.794 PM
LogName=System
EventCode=7036
EventType=4
ComputerName=WIN-71MP95N979J
Show all 12 lines
host = WebServer
source = WinEventLog:System
sourcetype = WinEventLog
Type here to search
9:28 AM
8/19/2026

Verification Testing

Switch to **Web Server** and log out of Web Server.

Log in to Web Server VM using invalid passwords two to three times in order to generate logs related to failed login attempts. Then, log in using the correct password

- Add extended filters to see the desired results



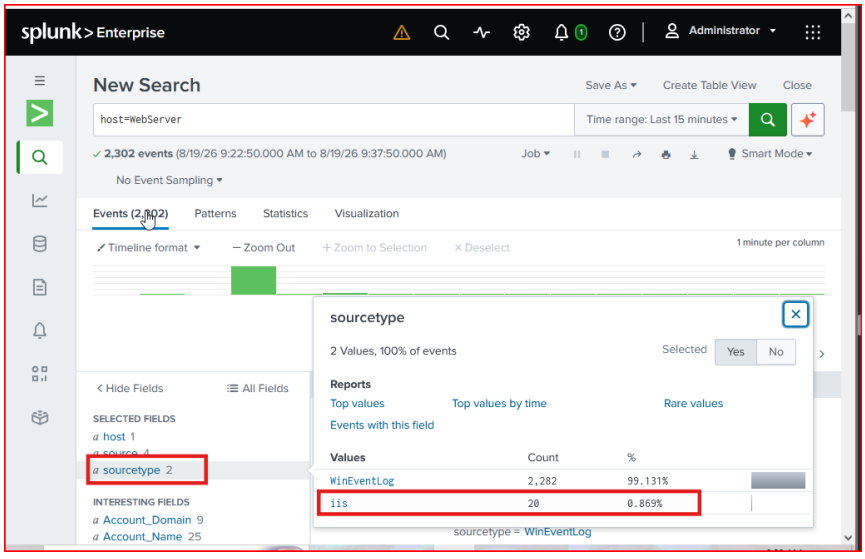
Generate IIS Activity

Go to **Attacker Machine** (192.168.100.143) or any machine

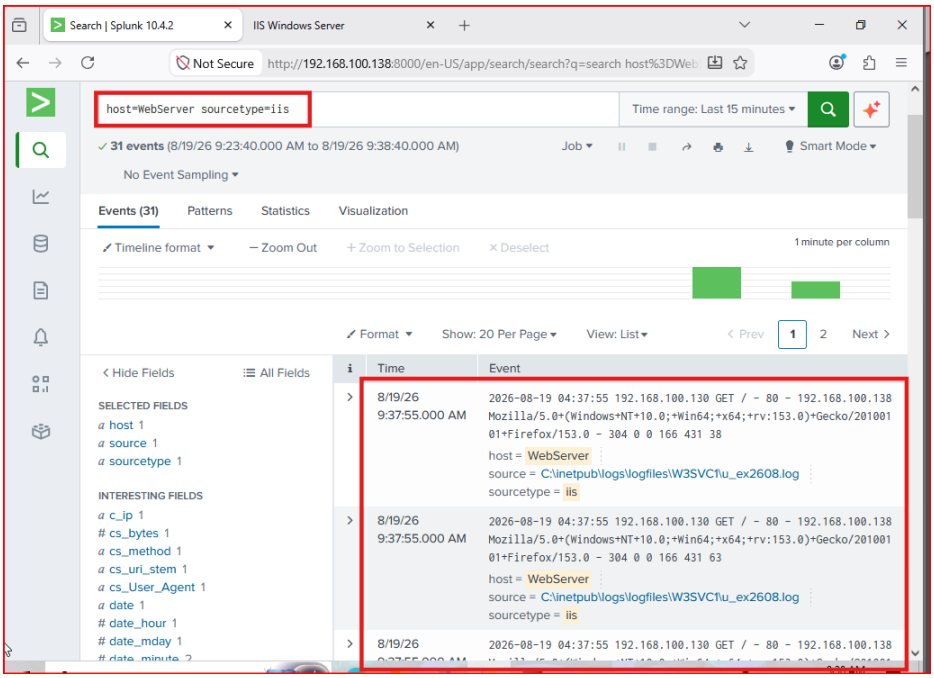
Open browser and visit: http://192.168.100.130

(Refresh page 5-10 times to generate traffic)

Then go back to Splunk: Select sourcetype -> iis



Showing new HTTP requests.



In this lab, Splunk Enterprise was successfully installed on the Windows 10 Admin Machine (192.168.100.138) as a centralized SIEM platform. The Splunk Universal Forwarder was installed on Windows Server 2019 (192.168.100.130) and configured to collect and forward security logs to the central Splunk instance.

Splunk Enterprise Installation

- Installed on Windows 10 Admin Machine (192.168.100.138)
- Web interface accessible on port 8000
- Receiving port configured on 9997

Universal Forwarder Deployment

- Installed on Windows Server 2019 (192.168.100.130)
- Successfully connected to Splunk receiver (port 9997)
- Connection verified with netstat (ESTABLISHED)

Log Collection & Forwarding

- Windows Event Logs successfully collected (2,415 events confirmed)
- Logs including:
    - Security events
    - System events
    - Application events
- All logs indexed in Splunk with sourcetype=WinEventLog

Centralized Log Analysis

- Search functionality working
- Can filter by Event ID, Log Name, Computer Name
- Timeline visualization showing log distribution
- Fields extracted for analysis (Account_Name, Account_Domain, ComputerName, EventCode, etc.)

Identifying Suspicious Activities Using Log Monitoring and Analysis

Use the already configured Splunk Enterprise (from Lab 3) to detect and analyze suspicious activities: brute-force login attempts (Windows Event Logs) and SQL injection attempts (IIS logs).

Objective

Generate brute-force failed logons on Web Server

Generate SQL injection attack on LuxuryTreats website

View and filter Windows Security logs (EventCode 4625) in Splunk

View and filter IIS logs for SQL injection patterns (cs_uri_query)

Confirm both attack types appear in the centralized Splunk console

Machines Required

Machine	IP / Name	Role
Admin Machine-1	192.168.100.138	Splunk Enterprise (central analysis)
Web Server	192.168.100.130	Target – Windows Event Logs + IIS logs
Attacker Machine	192.168.100.143	Generate SQL injection attack

Tools Required

Tool	Purpose
Splunk Enterprise	Centralized search & analysis
Splunk Universal Forwarder	Already forwarding logs from Web Server
Windows Event Viewer / Security Logs	Source of brute-force evidence
IIS Logs	Source of SQL injection evidence
Mozilla Firefox (Attacker)	Perform SQL injection

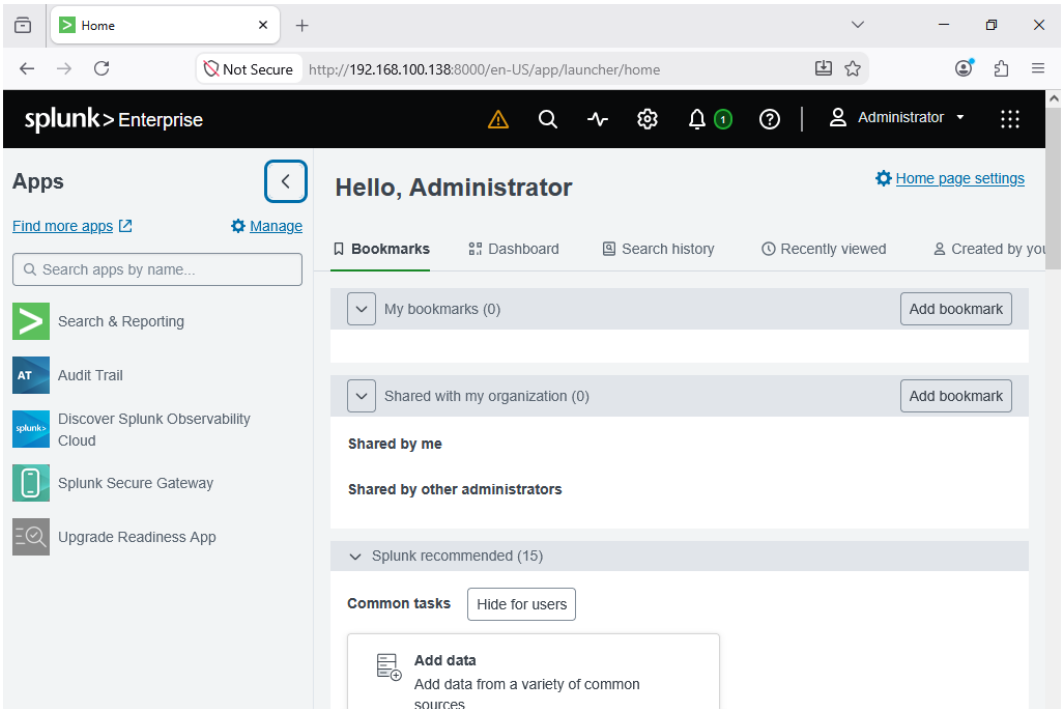
SplunkForwader status checking

1. Ensure **Splunkforwarder** service is running on Web Server (Services → **SplunkForwarder** → Start if needed).

```
PS C:\Users\Administrator> Get-Service SplunkForwarder

Status   Name               DisplayName
------   ----               -----------
Running  SplunkForwarder    SplunkForwarder
```

2. Confirm **Splunk Enterprise** is running on Admin Machine-1.



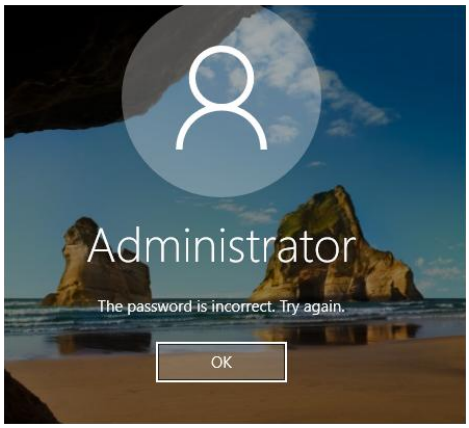
Machine: Web Server

Generating Brute-Force Activity

To perform the brute-force attempt on Web Server, launch Web Serve

Attempt three logins with the wrong passwords

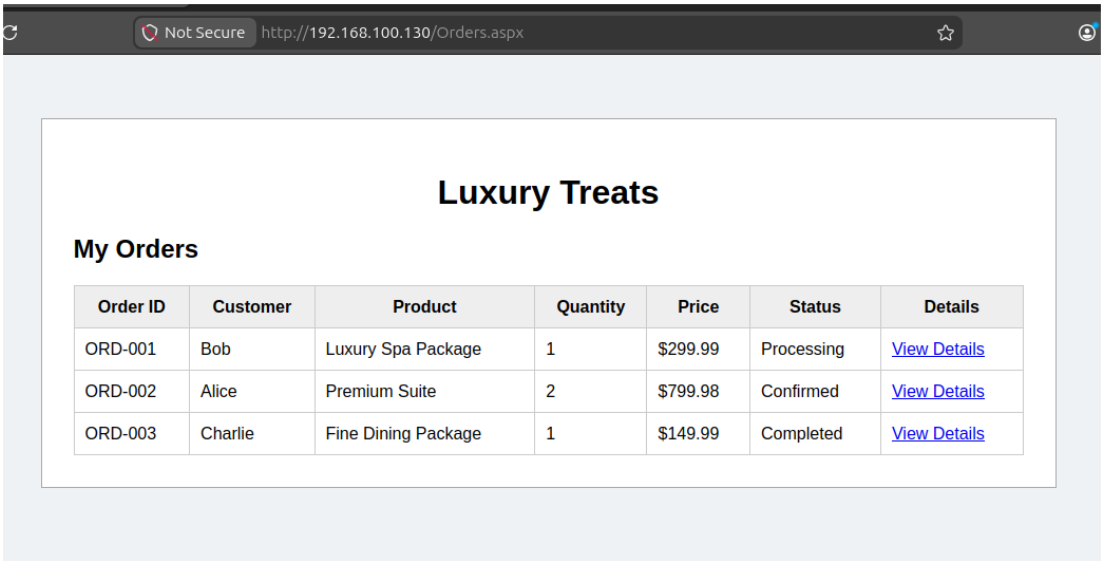
Next, log in with the correct password & leave the system running



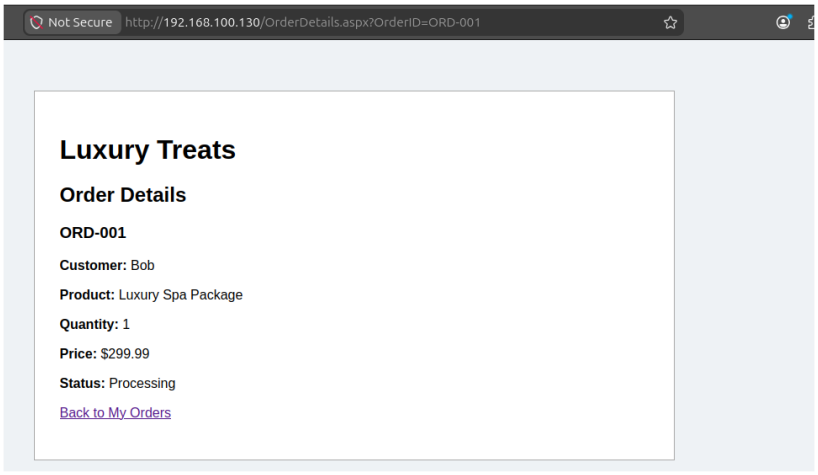
Generate SQL Injection Attack

Launch Attacker Machine

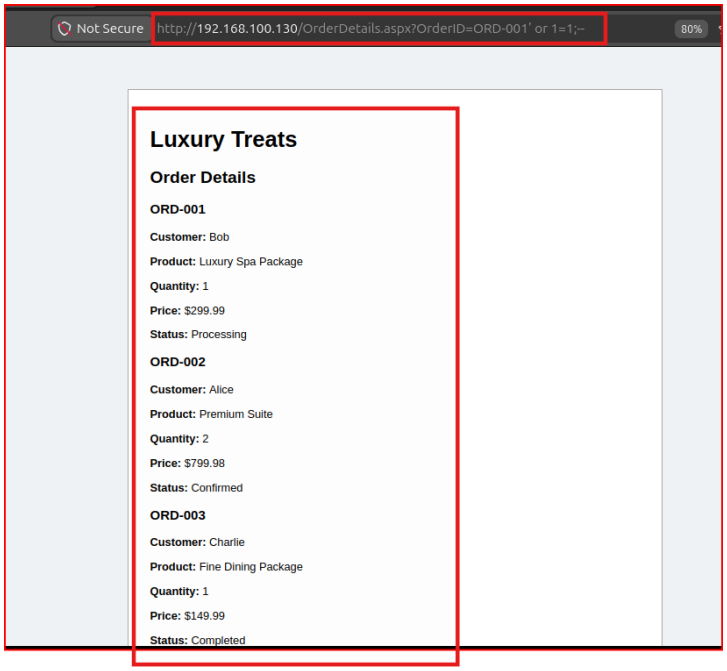
Open Firefox → go to http://192.168.100.130/Orders.aspx



Click **My Orders** → open Order Id **ORD-001**.




```
Modify the URL to: http://192.168.100.130/luxurytreats.aspx
/OrderDetail.aspx?id=ORD-001' or 1=1;--
```



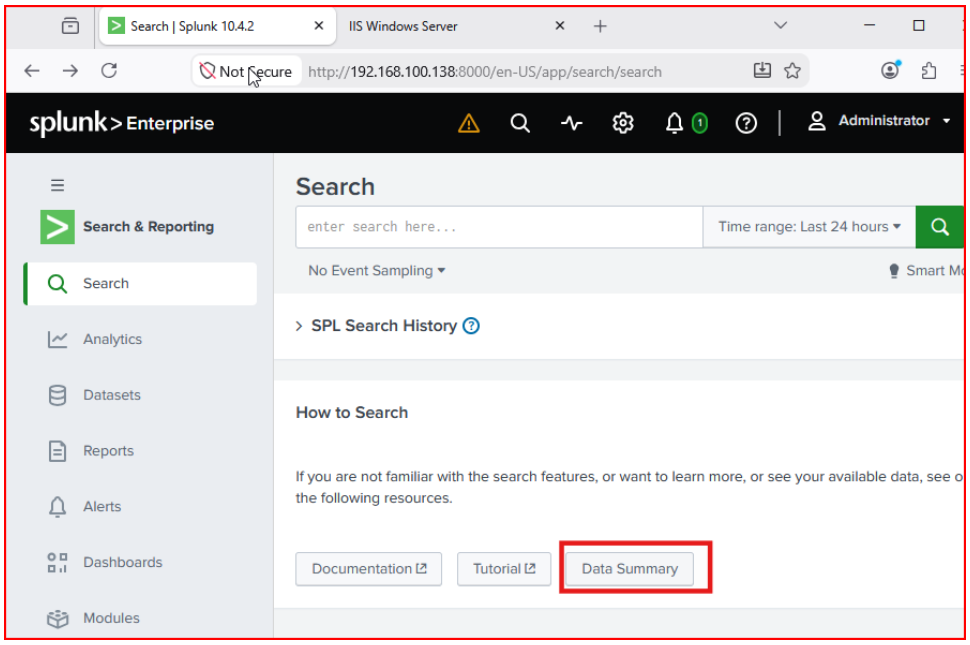
Press Enter. You will see other users' order details (SQL injection success).

Logout.

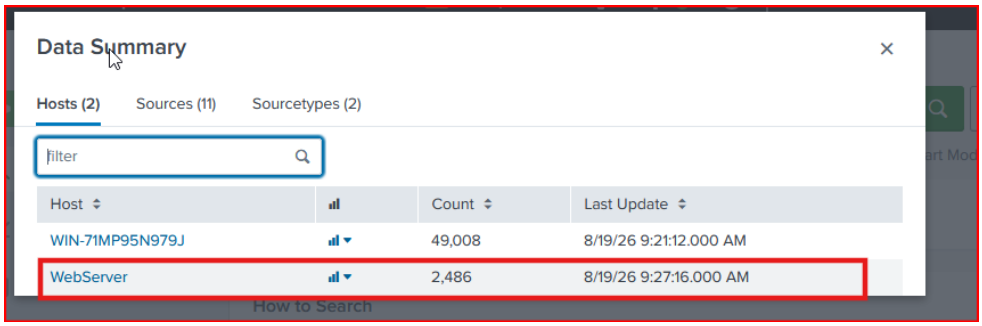
As per standard security practice, only the authorized user should be able to view their personal data. If a user can see other users' order details, then this can be considered a security breach, which is possible through an SQL injection attack. The attacker uses th1S technique to bypass the security measures protecting other users' data.

Analyze in Splunk (Admin Machine-1)

Go to Search & Reporting → Data Summary → select host WEBSERVER.

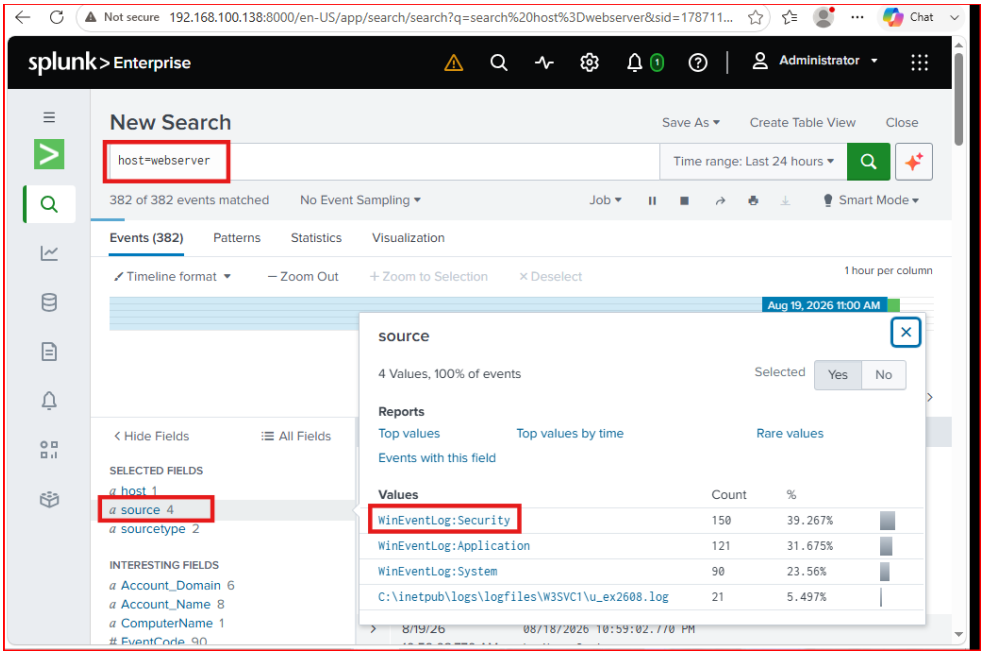


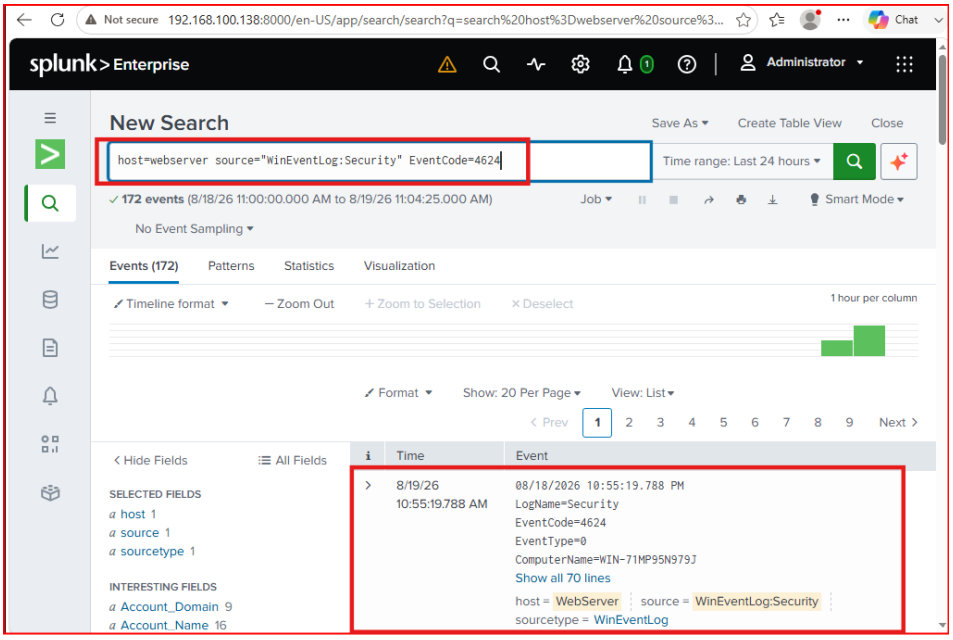
The host list will appear. Select the host machine (here, WebServer)



Detect Brute-Force

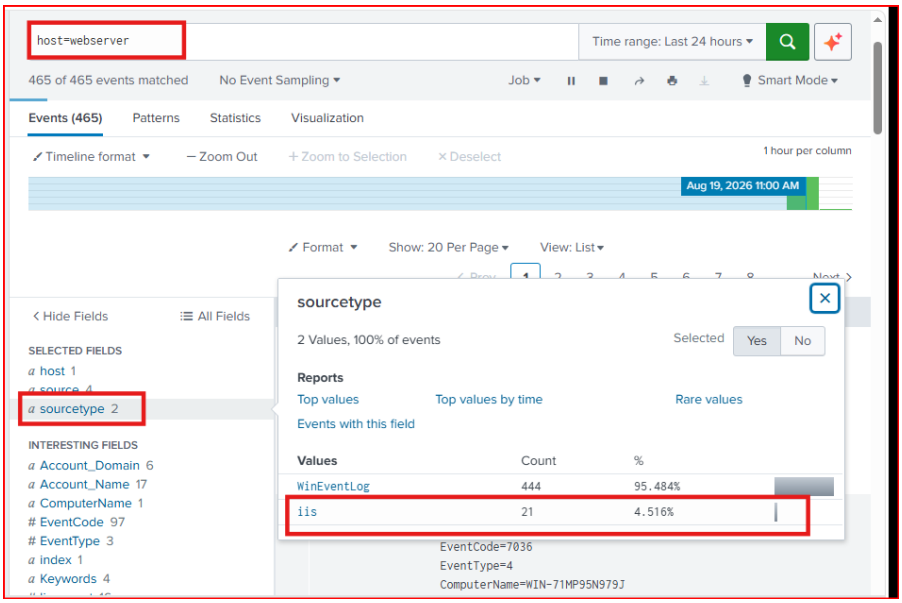
1. Click **sourcetype** → select WinEventLog:Security.
2. In search bar add: EventCode=4624
3. Press Enter. View failed login events. Click "Show all lines" on any event for details.





Detect SQL Injection

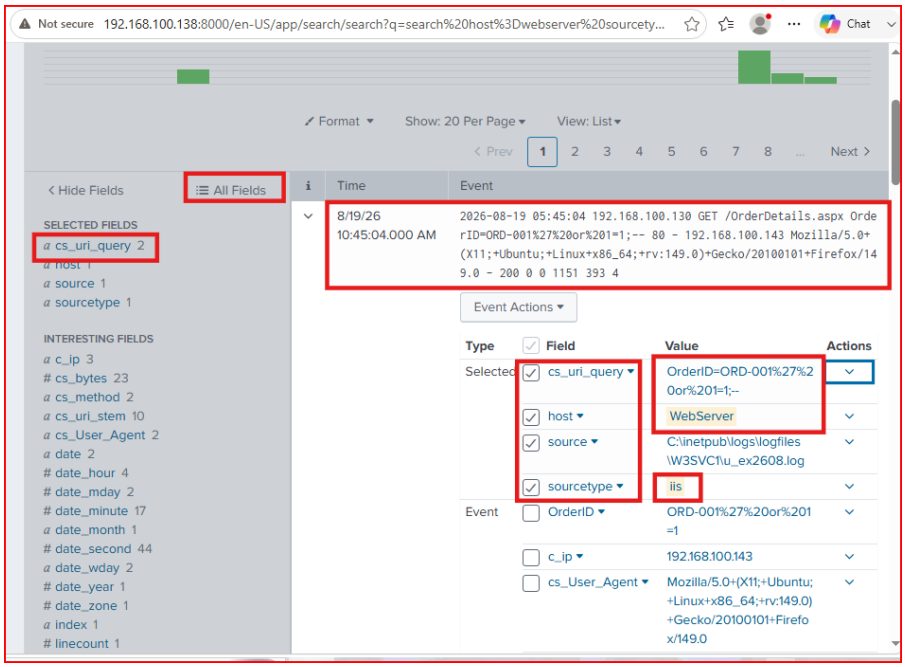
1. Clear previous search.
2. Click **sourcetype** → select iis.



3. Click **All Fields** → find and select cs_uri_query.



4. Click the cs_uri_query field → you will see the SQL injection string (' or 1=1;--).
5. Click the malicious value to filter and view the exact attack events.



Learned how a network defender uses Splunk to centrally identify real attack patterns (brute-force logons and SQL injection) that were generated against the Web Server. This demonstrates the power of centralized log monitoring for rapid threat detection.

Using New Relic for Network and System Log Monitoring

Install and use New Relic Infrastructure Agent on Admin Machine-1 to monitor system logs, logon events, and network traffic in real time (including detecting a ping flood / scan from the Attacker Machine).

Objective

Install New Relic Infrastructure Agent + Logs integration on Admin Machine-1

View real-time system logs and logon events

Monitor network transmit/receive traffic

Detect sudden traffic spike caused by ping from Attacker

Use "Diagnose with Lookout" and generate chart links

Machines Required

Machine	Role
Admin Machine-1	New Relic agent installation + monitoring
Attacker Machine	Generate network traffic (ping)

Tools Required

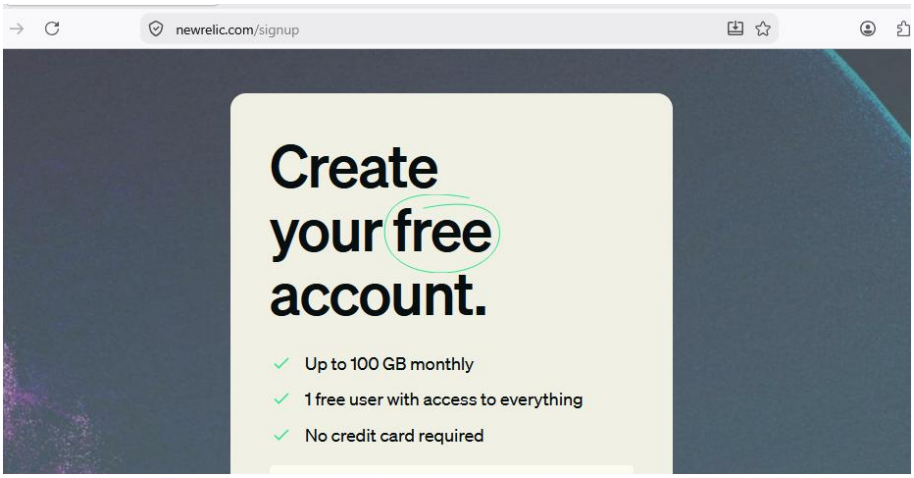
Tool	Purpose
New Relic (cloud account)	Real-time observability platform
New Relic Infrastructure Agent	Collects host, log, and network data
PowerShell (Admin)	Install the agent
Terminal (Attacker)	Generate ping traffic

Setup New Relic

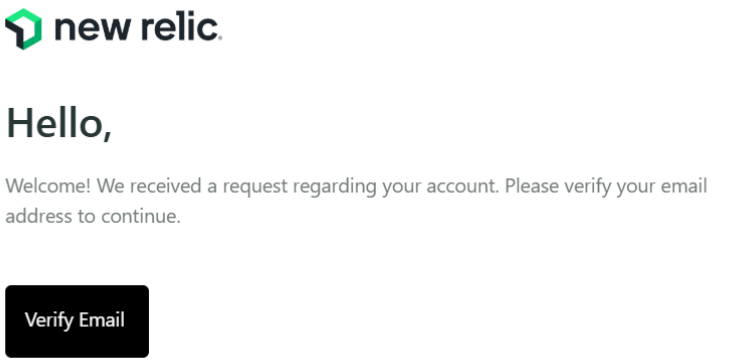
Login & Access New Relic

Navigate to https://newrelic.com

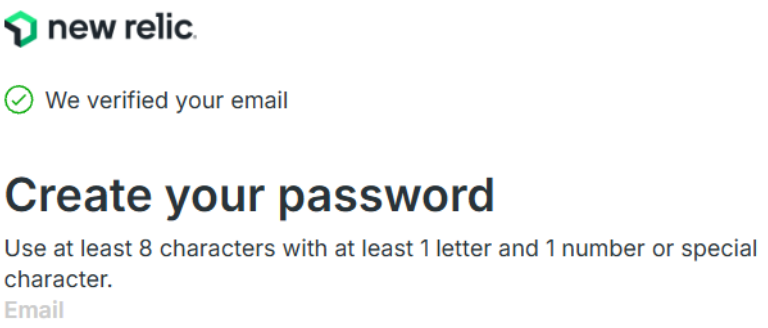
Create your account



Verify your mail



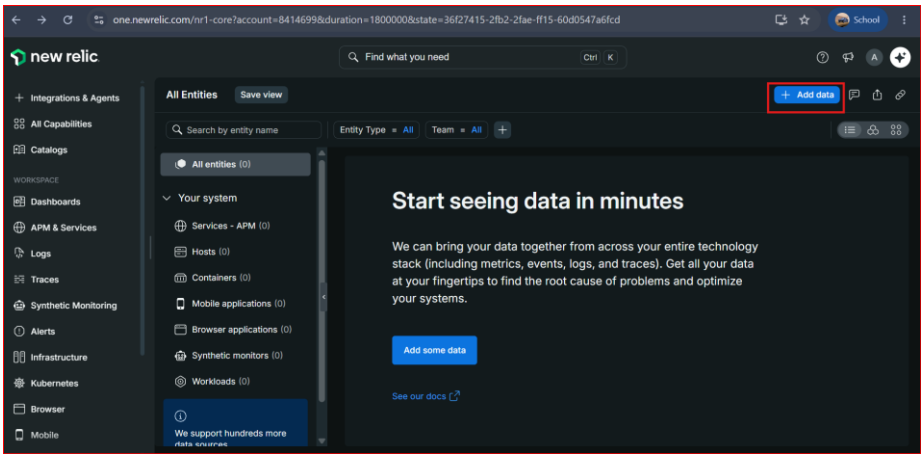
Setup your password



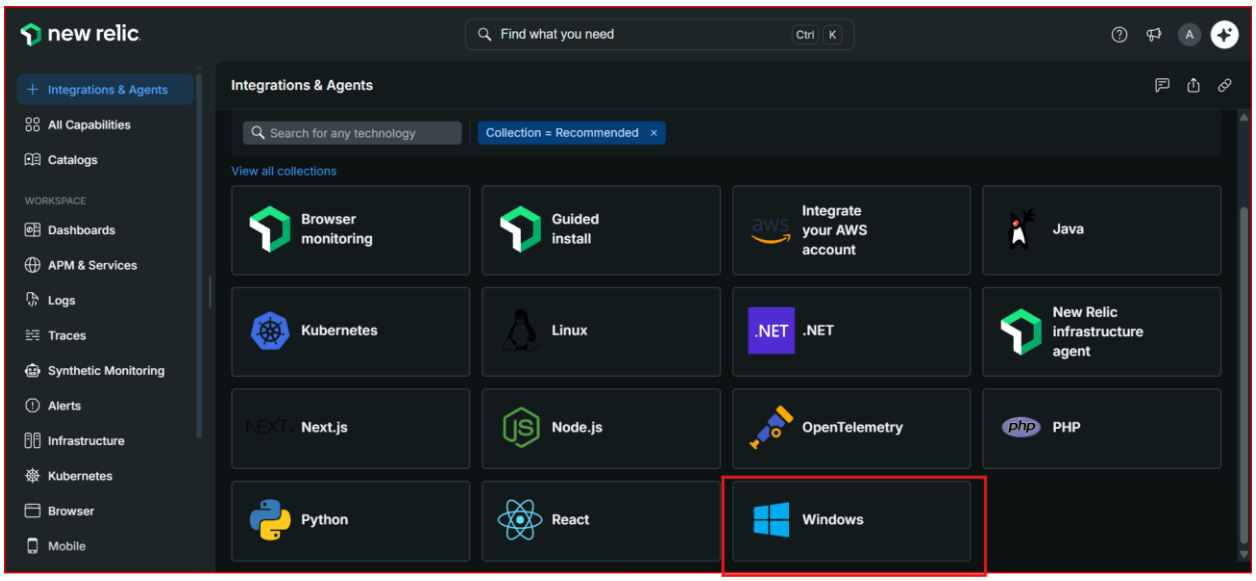
Open Chrome → go to https://newrelic.com → Log in with your registered New Relic account.

Install New Relic Agent

1. On Right top Select +Add data

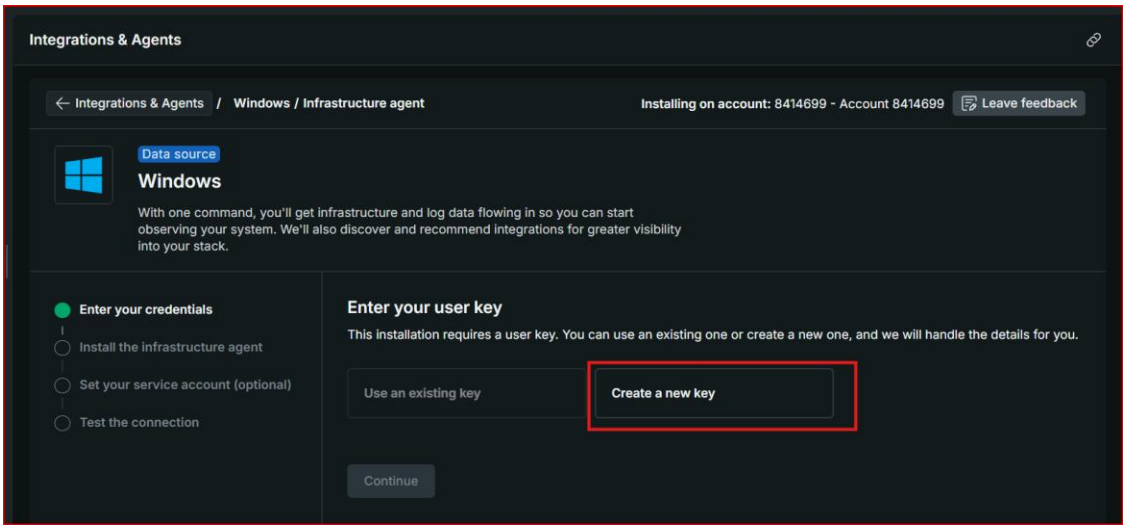


select environment **Windows**.

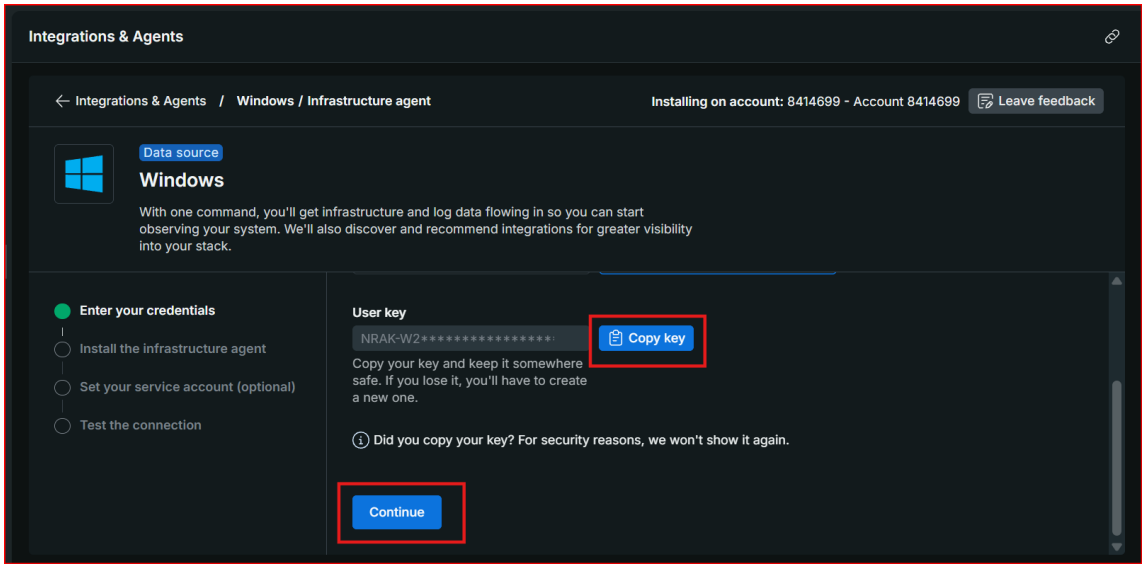


Create a new key

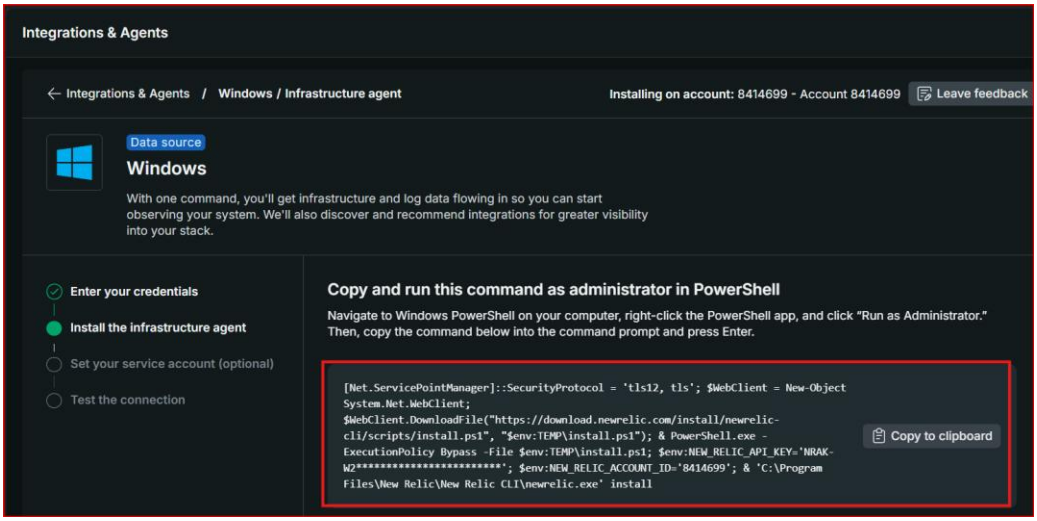
You don't need to manually generate anything. New Relic will create the key and handle it for the installation flow.



Save your key & Click Continue



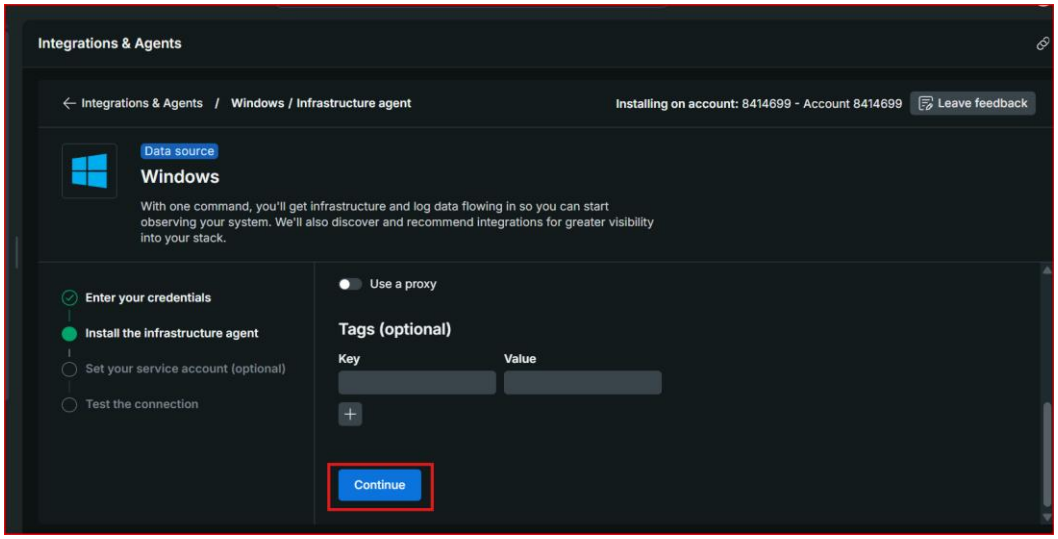
2. Copy the installation PowerShell command (Copy to clipboard).

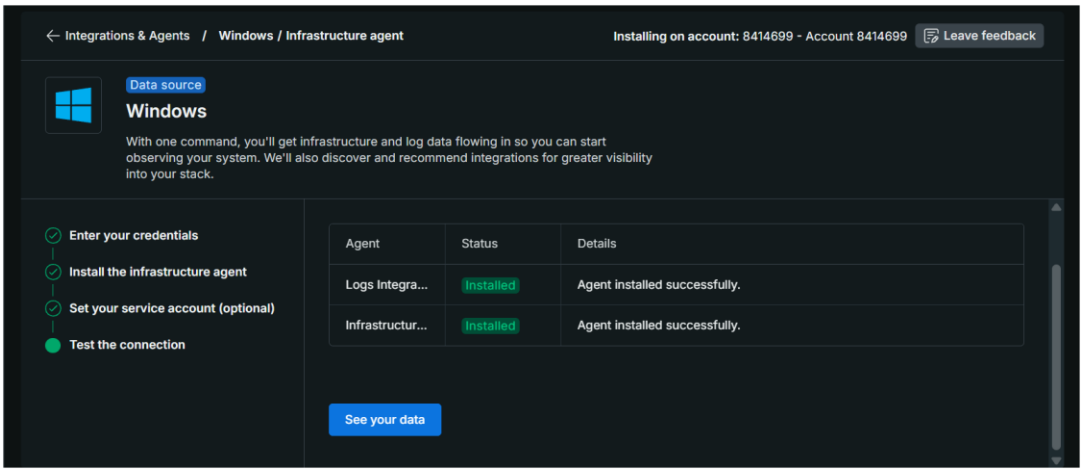


3. Open **PowerShell as Administrator**. **(Admin Machine)**
4. Paste the command and press Enter.
5. Wait until you see "New Relic installation complete" (Infrastructure Agent + Logs Integration installed).



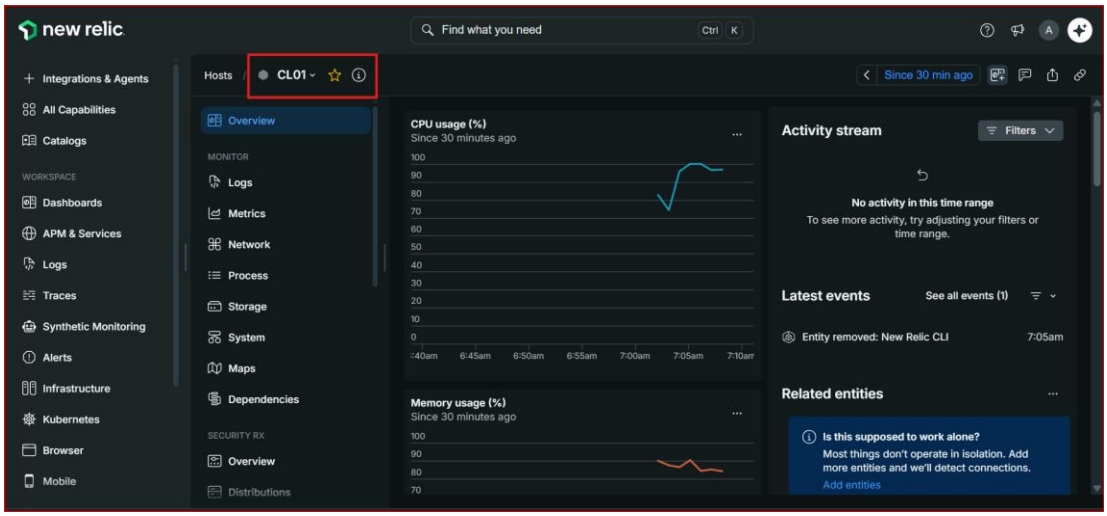
6. Go back to New Relic browser page → click **Continue** → **See your data**.





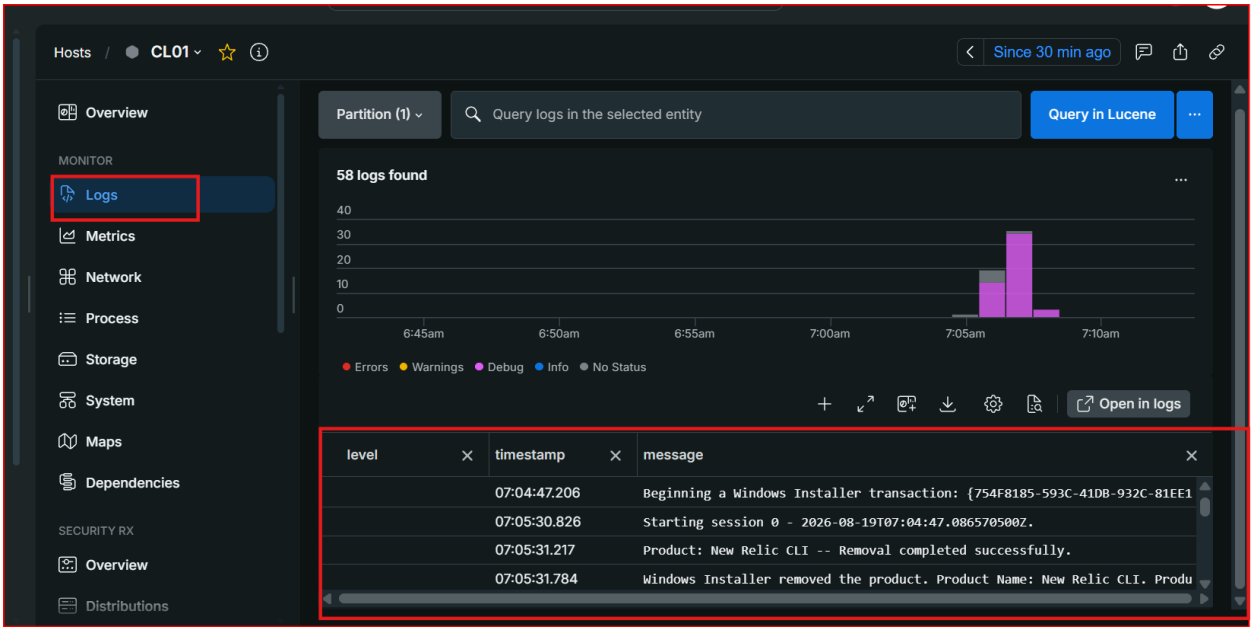
Explore Host Monitoring

1. You will see **Admin-Machine-1** under Hosts.
2. Left pane shows: Logs, Metrics, Network, Process, Storage, System + Diagnose with Lookout.



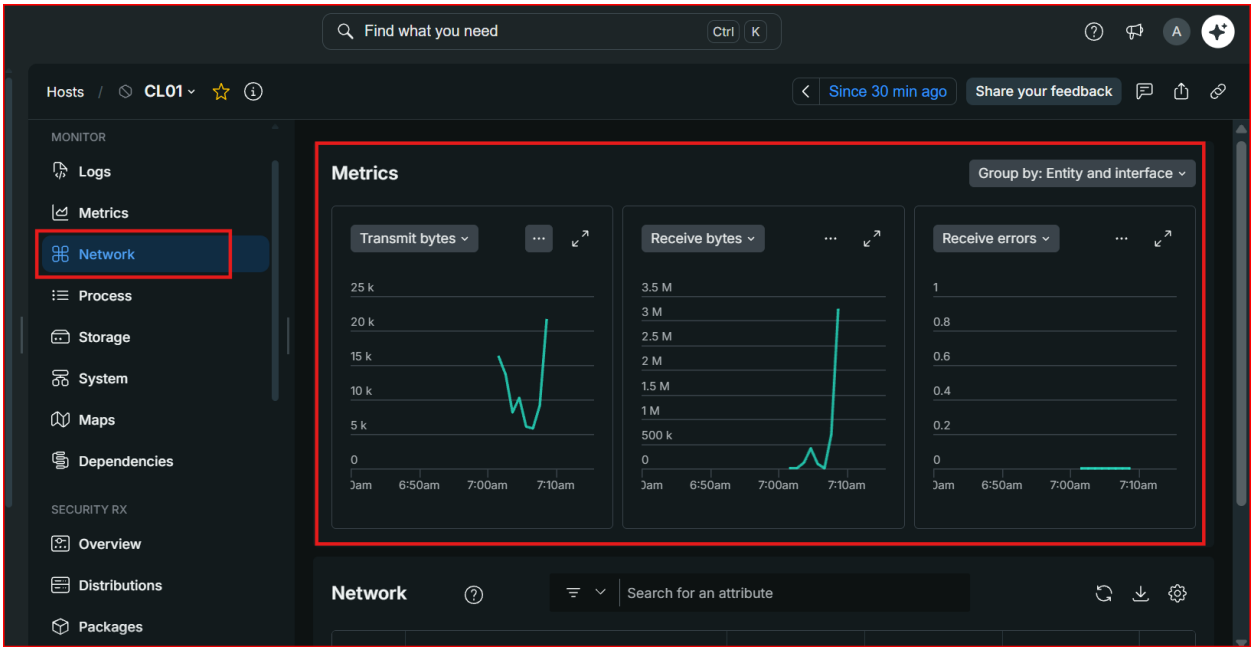
View System Logs & Logons

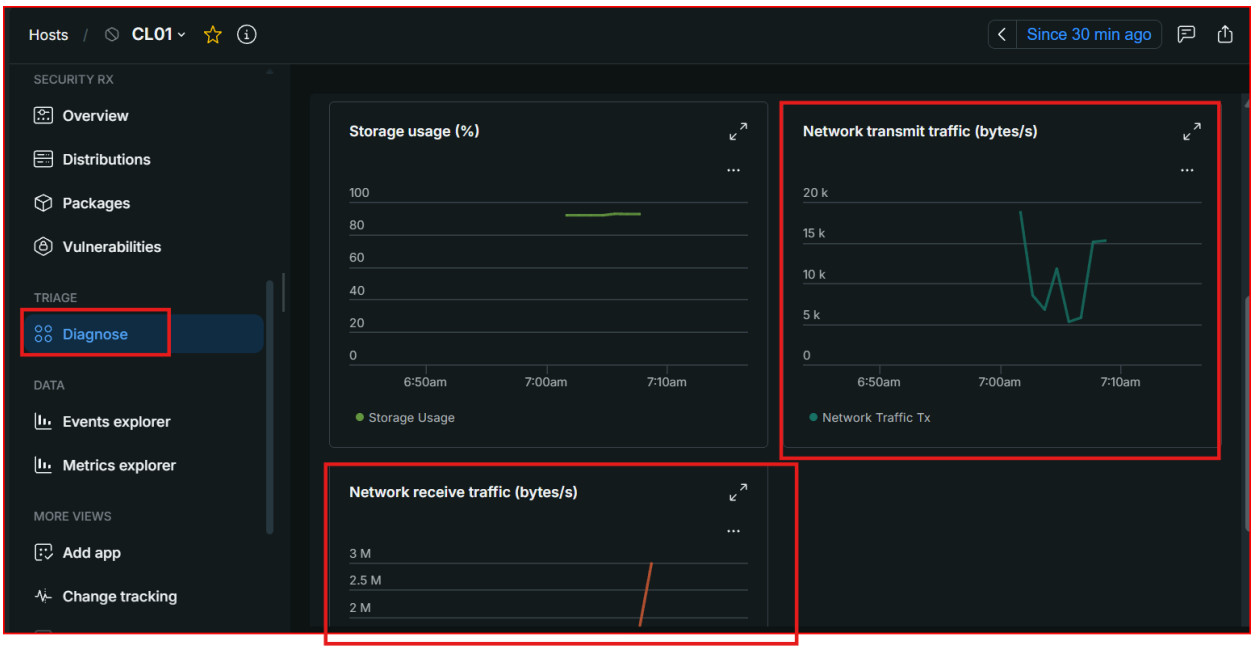
1. Click **Logs** under MONITOR.
2. Click **Open in logs**.
3. Observe timestamps and messages of successful logons and system events.

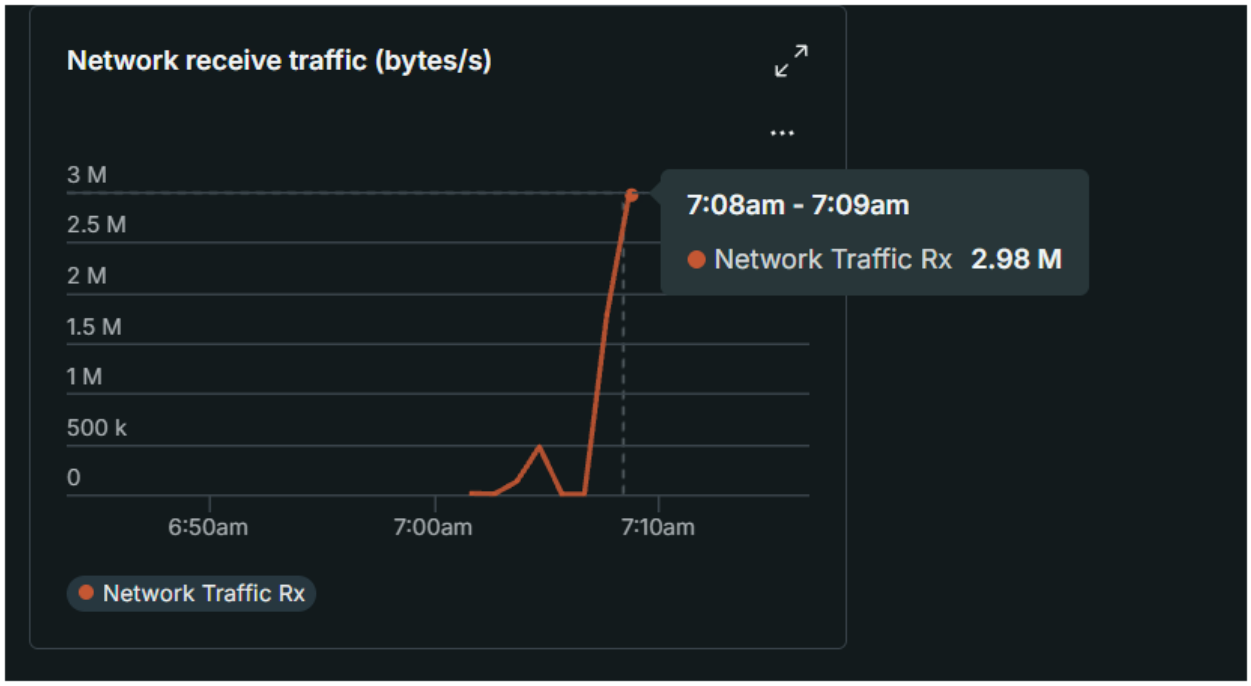


Monitor Network Traffic

1. Click **Network**.
2. Observe Transmit bytes / Receive bytes graphs.
3. Click **Diagnose with Lookout** → notice any high network transmit traffic.







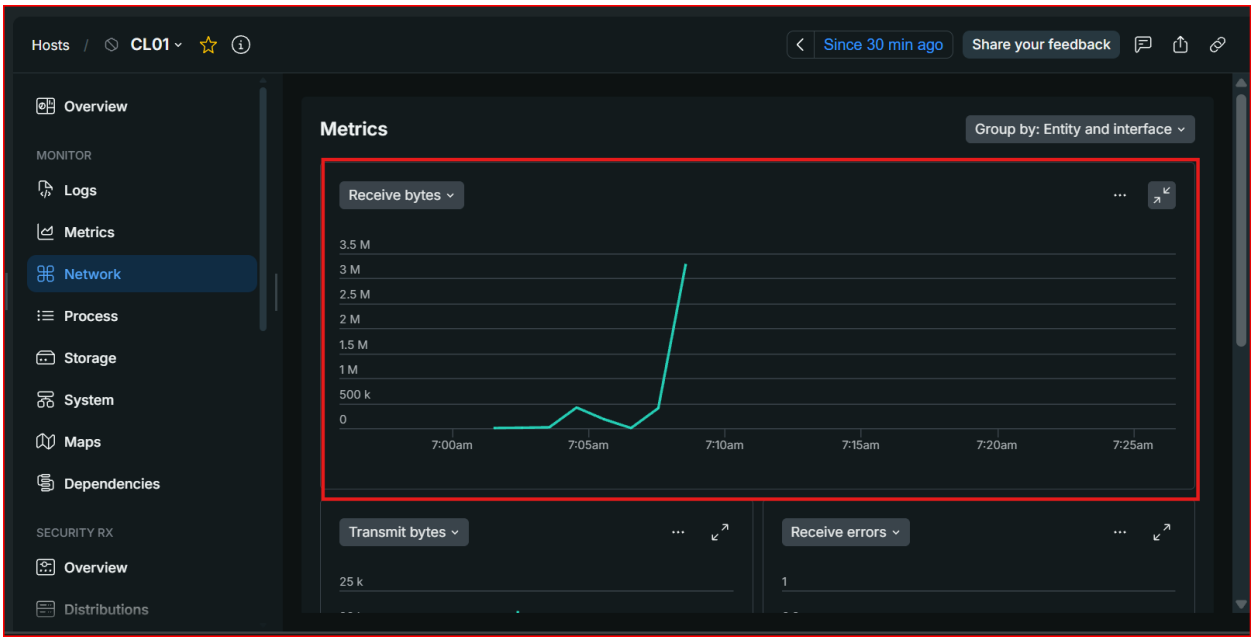
Generate & Detect Traffic Spike

1. Switch to **Attacker Machine**
2. Run: ping 192.168.100.138 (IP of Admin Machine-1). Let it run for a while, then stop.

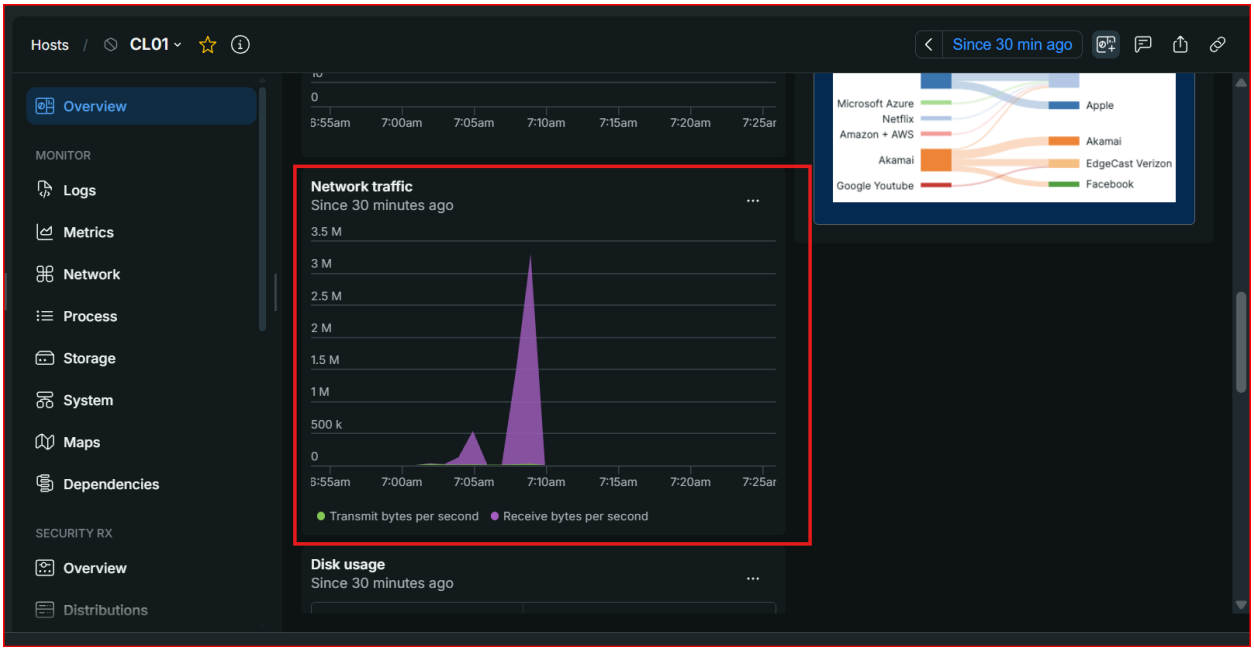
```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ ping 192.168.100.138
PING 192.168.100.138 (192.168.100.138) 56(84) bytes of data.
64 bytes from 192.168.100.138: icmp_seq=1 ttl=128 time=1.92 ms
64 bytes from 192.168.100.138: icmp_seq=2 ttl=128 time=0.559 ms
64 bytes from 192.168.100.138: icmp_seq=3 ttl=128 time=0.935 ms
64 bytes from 192.168.100.138: icmp_seq=4 ttl=128 time=0.639 ms
64 bytes from 192.168.100.138: icmp_seq=5 ttl=128 time=0.597 ms
```

3. Switch back to Admin Machine-1 → New Relic **Network** tab.

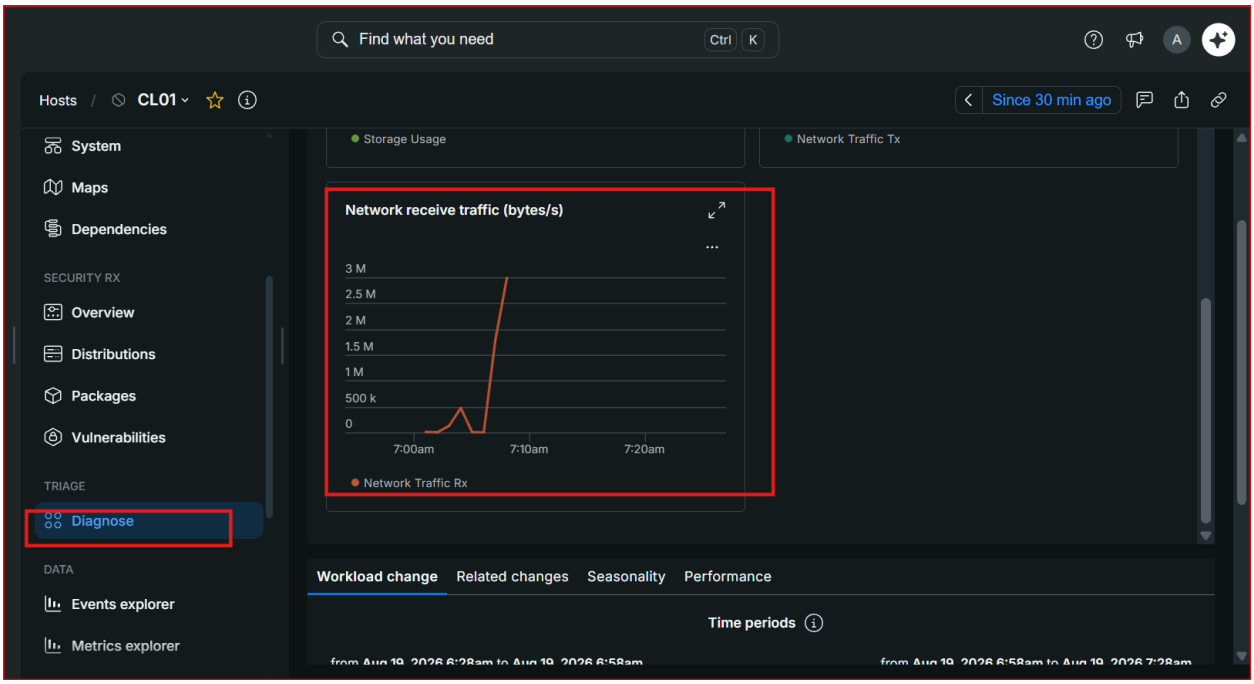
4. You will see a clear spike in Transmit/Receive bytes.



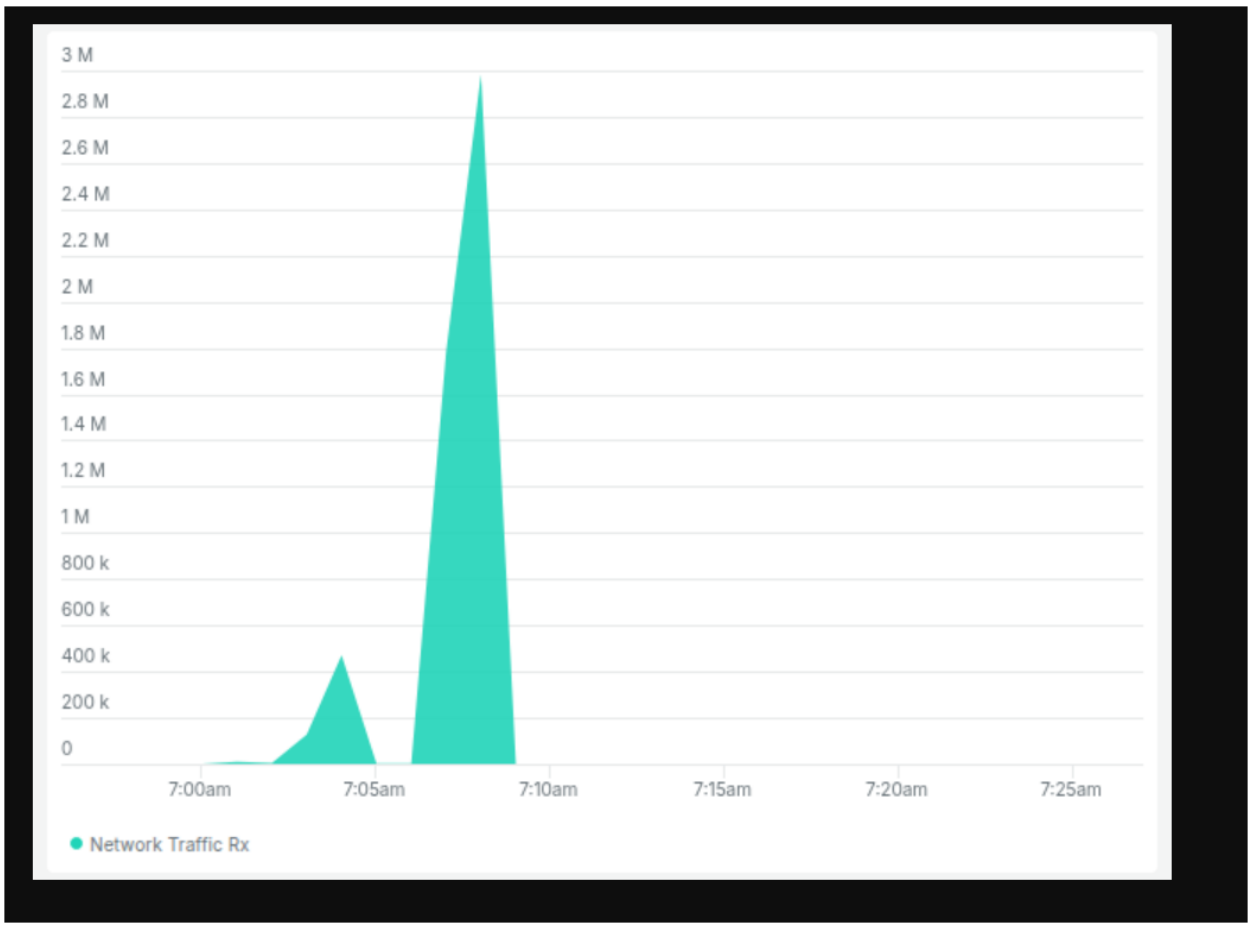
5. Go to **Summary** → Network traffic section (green graph = received bytes).



1. In **Diagnose with Lookout** → Network transmit traffic → use the three-dot menu → **Get chart link** → View / save the chart.
2. Repeat for Network receive traffic chart.







You successfully installed New Relic, monitored real-time system logs and logon activity, and detected a sudden network traffic increase caused by a ping scan. This demonstrates how modern observability tools help a network defender quickly identify anomalies and potential attacks.

Conclusion

From the first failed SSH attempt to the final New Relic spike, Module 15 delivered end-to-end log mastery.

Windows auditing was unlocked, Event IDs 4624 and 4625 captured and dissected, IIS access logs configured and examined, and every critical record successfully forwarded into Splunk for centralized search and correlation. New Relic then added real-time visibility, instantly flagging anomalous traffic and proving how modern observability tools accelerate detection.

The lab is complete. Local logs are no longer scattered, central visibility is operational, and anomalies can no longer hide. Defenders now own the story the network is writing.